



THE REPRODUCTIVE RESPONSE OF WILD
BOAR ON ENVIRONMENTAL AND
PHYSIOLOGICAL FACTORS AS WELL AS
HUNTING IN GERMANY



The reproductive response of wild boar on environmental and physiological factors as well as hunting in Germany

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Manuscript 1

The influence of environmental and physiological factors as well as hunting and population density on the litter size of wild boar (*Sus scrofa*) in an agriculture dominated area in Germany

Keywords: wild boar, reproduction, litter size, weather condition, population density, hunting, nutritional condition, mast years, agricultural intensification

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Abstract

The wild boar population has increased enormously in all of Europe over the last decades and caused problems like crop damage, transmission of diseases, and vehicle accidents. Therefore, it might be useful to investigate the underlying causes of this increase in order to be able to manage populations effectively. The purpose of this study was to analyse how environmental (food and climate) and physiological factors (maternal weight and age) as well as hunting influence the litter size of wild boar populations in Northern Germany.

During 12 hunting seasons (October-February) of data collection (2003-2014), 454 reproductive tracts of hunted pregnant (free-ranging) wild boar were sampled. During dissection of the reproductive tract, the number of foetuses was counted. The dressed weight of female wild boar was taken, and the age was examined by tooth determination. Meteorological data (temperature and precipitation per month), crop size of acorns and beechnuts as well as hunting index (HI=hunting bag/km²), which was used as an indication for population density, were available. Poisson distributed generalized linear models were analysed in R.

The mean litter size in the studied population for the whole period was 6.6 (range 1-12), which is one of the highest in all of Europe. Litter size was positively influenced by maternal condition (body weight and age) and it was larger under excellent mast conditions compared to moderate mast conditions or mast failure. However, no difference was found in the litter size under moderate conditions and in years of mast failure. Thus, a stable food supply by agricultural crops may buffer the negative impact of mast failures. Higher precipitation and temperature in autumn resulted in increased litter size so that weather conditions act indirectly on wild boar reproduction through condition and nutrition. Additionally, litter size increased with decreasing HI of the same year. This indicates that at high wild boar density, when the population is close to carrying capacity, density dependent juvenile recruitment is strong. A higher HI in the previous year did not affect the litter size, which demonstrates that hunters do not influence the litter size in wild boar. Also the following hunting year did not correlate with the litter size. This suggests that either hunters do not cull enough after a year of high litter size or piglets suffer from perinatal or postnatal mortality.

Introduction

Most ungulates are thought to be typical K-strategists characterized among others by a relative low litter size (Asher 2011; English et al. 2012).

Wild boar however, are a unique species showing traits from both K -and r-strategists. They show sexual size dimorphism (Focardi et al., 2008) and a low natural adult mortality rate in these climes (Keuling et al. 2013a), which is typical for K-strategists. On the other hand, they have by far the highest reproductive potential of all ungulate species world-wide in relation to its body mass (Boitani et al. 1995) so that they can reach high population densities within a very short time period (Geisser & Reyer, 2005). Additionally, it is also an opportunistic omnivore with a high adaptability to changing environmental conditions (Baubet et al. 2004; Schley and Roper 2003).

This possibility of varying between K -and r-strategist features depending on environmental conditions has caused an enormous increase of wild boar in the whole of Europe in the last decades (e.g. Keuling et al. 2013a; Lombardini et al. 2014; Massei et al. 2015). They can act as a reservoir of disease that can be transmitted to livestock (Gortázar et al. 2007). This problem becomes more urgent under high population densities (Cellina 2008). The spread of the African swine fever (ASF) in Europe among the dense wild boar populations is a recent risk (Sánchez-Vizcaíno et al. 2013) for animal welfare and economy (Mur et al. 2012). In agricultural dominated areas, wild boar can also cause enormous damage in crop fields (Geisser and Reyer 2005; Herrero et al. 2006; Keuling et al. 2014). Thus, it has become a very important species from an economical point of view (Kramer-Schadt et al. 2007; Uzal and Nores 2004) and it might be useful to investigate the underlying causes of this increase in order to be able to reduce wild boar density in the future by appropriate management (Keuling et al. 2008; Sodeikat and Pohlmeier 2007; Uzal and Nores 2004).

Factors determining various aspects of the reproductive biology are still the main area of research (Fernández-Llario et al. 2003) because it

directly affects population dynamics (Maillard and Fournier 2004). The litter size of wild boar is higher than in other similar-sized ungulates (Carranza 1996) and, even in comparison with several mammal species, wild boar belongs to the species with the highest litter size (Millar and Zammuto 1983). Therefore, it is an important parameter in the population growth of wild boar (Fernández-Llario and Mateos-Quesada 1998).

Several studies found that an increasing maternal weight or age (e.g. Aumaître et al. 1982; Fernández-Llario et al. 1999; Fonseca et al. 2011) increased the reproductive success with a larger number of fetuses per female.

Also environmental factors can influence the reproduction of wild boar (Bieber and Ruf 2005; Fernández-Llario and Mateos-Quesada 2005; Geisser and Reyer 2005). An increase in temperature caused by global warming in the last decades leads to a higher frequency of mast years in beech (Övergaard et al. 2007). As oak and beech are the most important natural sources of food for wild boar (Jędrzejewska et al. 1994), it results in large amounts of food (Jędrzejewska et al. 1997) and higher maternal body weight that increases the energy to invest in offspring (Bieber and Ruf 2005; Geisser and Reyer 2005). At the same time, intensification of agriculture leads to an all year round food availability (Massei et al. 1996; Schley and Roper 2003).

Also hunting (or population density) might have an effect on the litter size production (Massei et al. 1996; Servanty et al. 2011). Due to their large size and long life span, normally, ungulate populations are close to carrying capacity (McCullough 1999). However, if population density cycles, females produce fewer high quality offspring at peak density (Sinervo et al. 2000) so that we can expect a lower litter size when wild boar density is high (Massei et al. 1996).

Even if some studies about environmental effects (Bieber and Ruf 2005; Fernández-Llario and Mateos-Quesada 2005; Geisser and Reyer 2005; Servanty et al. 2009), physiological effects (Fernández-Llario et al. 1999; Fonseca et al. 2011; Sáez-Royuela and Tellería 1987) and hunting (Servanty et al. 2009) on the reproduction of wild boar have already been published in Europe, only few researches show results over a

long time period in an agriculture dominated area. Additionally, none of them focuses on so many different influencing parameters in one study, which is important to get a broad insight in the factors affecting reproduction in wild boar.

The purpose of this study was to analyse how environmental (food and climate) and physiological factors (maternal weight and age) as well as hunting influence the litter size of wild boar populations in Germany.

Material and methods

Study area

The study area is situated in the east of the federal state of Lower Saxony (Northern Germany) (52.36°N, 10.35°E) (Appendix I) where altitude ranges from 60 to 130 m ASL (Gethöffer 2005). Based on data from the German Meteorological Service, the average annual temperature from 2000 until 2010 was 9.7°C (DWD 2015). The average annual precipitation per year was 807 mm from 2000 until 2010 (DWD 2015).

The size of the study area is about 4885 km². Agriculture and forest comprise 54.6 % and 27.6 % of the area, respectively. The remaining 17.8 % are buildings and infrastructure (NLS 2007).

The agricultural land is characterized by 43.3 % grain (oat, barley, rye, and wheat), 16.0 % grassland, 11.7 % sugar beets, 6.6 % oilseed rape, 6.0 % maize, and 5.8 % potatoes. The remaining 10.6 % consists of heathland, fallow land and horticultural products (NLS 2007).

The forest consists of 18.9 % oak (*Quercus* spp.), 11.2 % beech (*Fagus sylvatica*), 7.6 % other deciduous trees (e.g. *Betula* spp, *Acer* spp., *Tilia* spp., *Sorbus aucuparia*), 45.5 % pine (*Pinus sylvestris* and *P. strobus*) and 16.6 % other coniferous trees (*Larix* spp., *Pseudotsuga menziesii*, *Picea abies*).

Supplementary feeding is not allowed in the study area. However, baiting in the winter with the help of maize is a necessary hunting strategy (Keuling 2015). Approximately not more than 1000 kg supplemental food per 100 ha is supplied every year for baiting purposes (similar to Keuling et al. 2008), whereof about 200 kg are actually eaten by wild boar (Keuling 2015).

Data collection

In total 3011 reproductive tracts (uteri and ovaries) were collected from free-ranging hunted wild boar sows in the study area in the hunting years 2003 until 2014. 454 wild boar sows were pregnant and their reproductive tracts were used for analysis. Wild boar were sampled from October until February (Appendix II). Hunting season 2009, for instance, starts in October 2009 and goes until February 2010. Each sampled individual is associated with a locality and thus with one of the nine administrative districts (Appendix I). Dressed weight (without digestive system, heart, lungs, liver, reproductive tract and blood) of female wild boar was taken. The age of the wild boar sows was identified by tooth determination (Gethöffer et al. 2007). The lower jaw of the sampled individuals was examined, and the tooth eruption, replacement, and dental abrasion were assessed. Based on this age identification, wild boar sows were pooled in three classes (piglets: ≤12 months, yearlings: ≥13-24 months, adults: ≥25 month). (Appendix II). During dissection in the laboratory, the number of foetuses was counted.

Meteorological data (temperature and precipitation per month) were supplied by local meteorological services (DWD 2015). The mean temperature and precipitation of the two seasons, summer (June, July, August) and autumn (September, October, November), before/during data collection were calculated. Each sampled individual is associated with an average temperature and precipitation per month in summer and autumn. The associated weather parameter depends on the hunting year and administrative district where the individual sample was collected. Each administrative district was associated with an appropriate weather station close by.

Summer climate data indicate dry summers in 2003, 2009 and 2013, whereas the summers of 2004, 2007 and 2011 were relatively wet (Appendix III). The summers were relatively warm in 2003, 2006 and 2010 but cold in 2005 and 2012 (Appendix III).

In 2006, 2011 and 2014, the amount of mean autumn precipitation was low whereas it was high in 2007, 2009 and 2010 (Appendix IV). The years 2003, 2007 and 2010 had a relatively cold autumn

whereas in 2006 the mean autumn temperature was quite high (Appendix IV).

Data on the amounts of acorn and beechnut crop size were provided by The Forestry Seeds Advisory Board (FSB) at Oerrel/Lower Saxony, Germany. The yield values of beech (*Fagus sylvatica*) and oak (*Quercus robur* and *Q. petraea*) per year (from 1 July until 30 June) for Lower Saxony were transformed in three classes indicating the condition of the mast crop. Excellent mast crop demonstrates > 70 % of the maximum possible harvest, moderate mast crop condition indicates 10-69 % of the maximum possible harvest, and failure is defined as < 10 % of the maximum possible harvest (Keuling et al. 2013b). Mast crop of beechnut was excellent in the years 2004 and 2009 whereas 2005, 2008, 2010, 2012 and 2013 were years of mast crop failure for beechnuts (Appendix V). Mast crop conditions for oaks were excellent in 2003, 2006 and 2009 whereas 2004, 2008, 2010 and 2014 were years of failure in acorns (Appendix V).

Each sampled individual is also associated with a hunting index (HI) of the administrative district where it was collected (Appendix I). The HI is defined as the hunting bag per km² and as data were collected mainly with the same hunting technique, we expect that the effort was the same in each year. In 2002, the HI was at about 2.0 and decreased in 2003 to 1.5, where it stayed at that level until 2005. In 2006, it decreased further to a value of about 1.2. Until 2008 it increased again and reached a higher level than in 2002 (2.2). It stayed around that level until 2011, where it dropped to 1.5, followed by a peak of almost 2.0

in 2012 and a low level of 1.5 in 2013 again (Appendix VI).

To be able to analyse the effect of wild boar density on the litter size, estimations of the abundance of wild boar in the study area are necessary. Unfortunately, the expense and effort required to estimate true population densities are prohibitive (Acevedo et al., 2007). Data required by trap cam method (Rowcliffe et al. 2008) are only partly available for the study area so that the HI has to be taken as an indication for the population density of wild boar. The HI is regarded as an appropriate measurement to obtain valid results about the wild boar density (Acevedo et al. 2007).

Statistical analysis

For all tests, significance was assumed when $p < 0.05$. All analyses were performed using R software version 3.1.2 (R Core Team 2014).

Since there were correlations between the three different parameters (food, weather, and hunting), and because we are interested in the effect of all three parameters on the litter size, three poisson-distributed generalized linear models were performed.

Maternal body weight and age correlated with each other and could not be put together in one model. Because we expected that they might interact with weather and mast conditions, body weight and age class were included in the model with mast and weather condition, respectively. The Akaike Information Criterion (AIC) (Burnham and Anderson 2002) was used to decide which physiological parameter should be

Table 1: GZLM including weather condition and age class of the 12 wild boar hunting years in the study area (n=454). Piglets are used as reference group (n=119) for the age class. Precipitation and temperature are for the autumn preceding data collection of the same hunting year. Estimates, standard error (SE) and z-values of the best fitting model are shown. Significance is indicated in the following way: * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$.

Variable	Estimate	SE	z value	Significance	Effect
Yearlings (n=209)	0.330228	0.031888	10.356	***	Higher litter size compared to piglets
Adults (n=126)	0.422760	0.034119	12.391	***	Higher litter size compared to piglets
Temperature in autumn (n=454)	0.036650	0.013342	2.747	**	Increased litter size with higher temperature
Precipitation in autumn (n=454)	0.001933	0.000857	2.256	*	Increased litter size with higher precipitation

included in which model (mast or weather conditions). The last model included the HI from the same hunting year where samples were collected, the HI from the previous hunting year compared to the year where samples were collected, and the HI from the following hunting year compared to the year where samples were collected. For all models, interaction terms were tested, but are not reported unless statistically significant.

Explanatory variables were checked for correlations between each other. If a Pearson correlation was larger than 0.7, one of the two variables was excluded from analysis (Zuur et al. 2010). Models were validated by checking Cook's distance and temporal autocorrelation.

Pearson dispersion statistic was under 1 indicating underdispersion (Hilbe 2014). VGAM package in R was used to correct for underdispersed data (Yee 2010).

AIC was used to drop each explanatory variable in turn until the best fitting model was reached.

Results

The mean litter size in the studied population for the whole period was 6.6 ($\pm$ 2.0 SD; $n=454$) ranging from 1 to 12 fetuses per sow. In the best fitting model including weather conditions and age class (Appendix VII), all three age classes differed highly significantly from each other (Table 1).

Mean litter size was highest in adults (7.5 fetuses per female sow) and lowest in juveniles (4.9 fetuses per female sow). Higher precipitation and temperature in autumn during data collection also resulted in a larger litter size (Table 1).

Modelling mast crop of acorns and beechnuts and maternal weight (Appendix VII) revealed

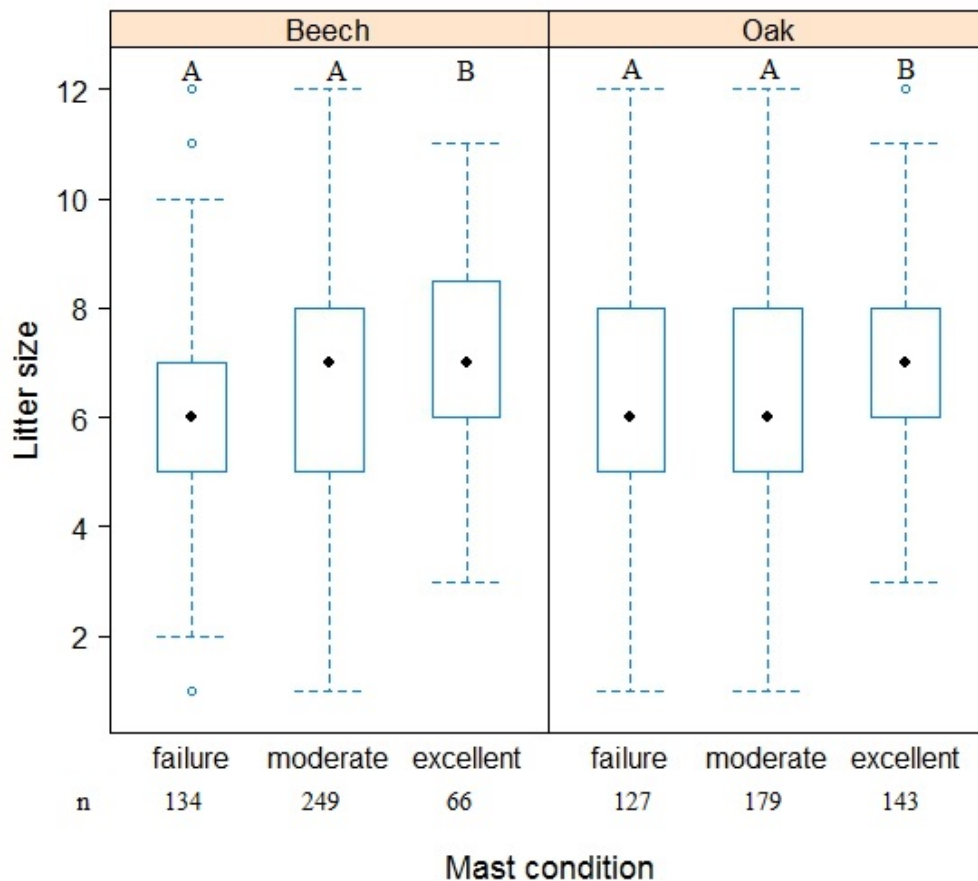


Figure 1: Litter size per mast crop condition of oak (*Quercus robur* and *Q. petraea*) and beech (*Fagus sylvatica*) of 12 wild boar hunting years in the study area ($n=449$). Box and whisker plots show median (point within box), 25% (interquartile range 1) and 75% (interquartile range 3) percentiles (box) and the range (whiskers), which includes observations up to 1.5 times the interquartile ranges. Circles indicate statistical outliers (observations outside 1.5 times the interquartile ranges). A is significantly different ($p<0.05$) from B. n indicates the sample size per mast crop condition and tree species.

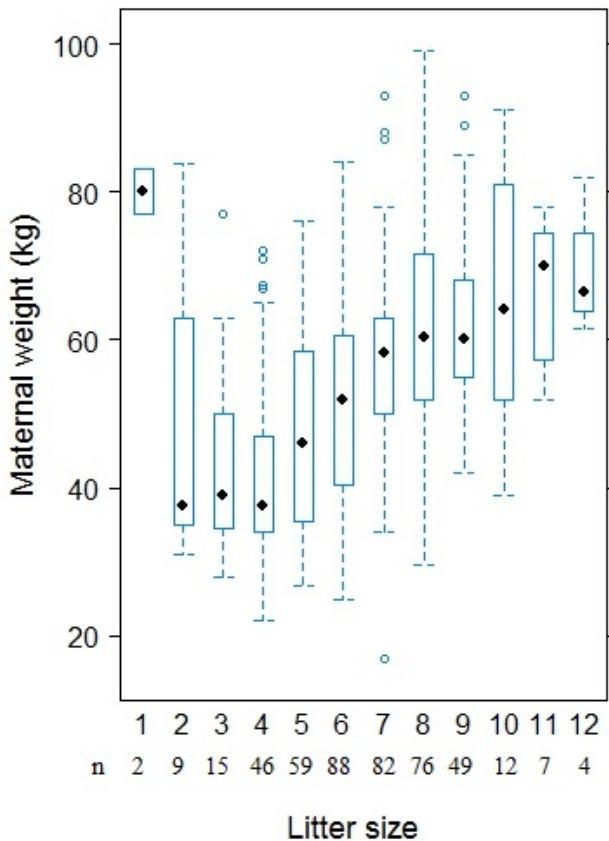


Figure 2: Maternal body weight in kg per litter size of 12 wild boar hunting seasons in the study area. n indicates the number of samples per litter size. Box and whisker plots show median (point within box), 25% (interquartile range 1) and 75% (interquartile range 3) percentiles (box) and the range (whiskers), which includes observations up to 1.5 times the interquartile ranges. Circles indicate statistical outliers (observations outside 1.5 times the interquartile ranges).

significantly higher litter size in excellent mast crop conditions compared to the litter size in years of mast crop failure in oak as well as in beech (GZLM, oak mast crop failure: $z=-2.917$, $n=127$, $p<0.01$; beech mast crop failure: $z=-2.356$, $n=134$, $p<0.05$). In oak, excellent mast crop conditions led to significantly higher litter size than under moderate mast crop conditions or failure mast years (GZLM, oak failure mast crop condition: $z=-2.917$, $n=127$, $p<0.01$; oak moderate mast crop condition: $z=-3.359$, $n=179$, $p<0.001$). In beech, under excellent mast crop conditions, litter size was only weakly significantly higher than litter size under moderate mast crop conditions (Figure 1) (GZLM, beech moderate mast crop condition: $z=-1.653$, $n=249$, $p<0.1$). Also higher maternal weight resulted in greater litter size (Figure 2) (GZLM, $z=11.574$, $n=449$, $p<0.001$).

The best fitting model concerning the HI

(Appendix VII) indicates that decreasing HI demonstrated a significantly higher litter size of the sows from the same year (GZLM, $z=-2.123$, $n=395$, $p<0.05$). However, the litter size of the following as well as the previous hunting year did not show a significant difference with increasing or decreasing HI (following hunting year: GZLM, $z=1.082$, $n=395$, $p=0.2794$; previous hunting year: GZLM, $z=1.542$, $n=395$, $p=0.1230$).

Discussion

Results show that litter size was influenced most positively by physical maternal conditions (age and weight). Also environmental parameters affected wild boar litter size. At excellent mast conditions of oak and beech, we found a higher litter size of wild boar sows compared to failure or moderate mast crop conditions. Higher precipitation and temperature in autumn resulted in increased litter size. Litter size increased with a decreasing HI of the same year. However, the HI of the following year as well as the previous year did not correlate with the litter size.

A litter size of 6.6 fetuses per female wild boar is recorded to be one of the highest in Europe (Fonseca et al. 2011).

In Southern Europe (Fernández-Llario and Mateos-Quesada 2005; Lombardini et al. 2014; Rosell et al. 2012; Servanty et al. 2007), in Northern Europe (unpublished, Malmsten et al.) as well as in mountainous regions litter sizes are lower (Moretti 1995). Thus, those findings are not in accordance with the pattern described by Bywater et al. (2010), who documented a strong positive effect of latitude on mean litter size of wild boar in Europe. A possible reason could be that different environmental conditions (fewer food resources and harsher (colder or drier) weather conditions) have a negative effect on the litter size in Northern and Southern Europe. Also litter size ranging up to more than 9 offspring was only described by Servanty et al. (2007) in a French study area.

Our results are supported by several studies where a positive correlation of body weight and litter size was found (Boitani et al. 1995; Fernández-Llario et al. 1999; Fernández-Llario and Mateos-Quesada 1998; Fonseca et al. 2011;

Rosell et al. 2012).

During the 12-year study period, average dressed maternal body weight increased in piglets, yearlings, and adults with 12 %, 13 %, and 11 %, respectively (Lustig 2015), which should be seen as a sign of caution that litter size might increase in the future when weight increases. This statement is supported by the results documented by Apelius (1995) who found an average litter size of 4.67 and 5.11 for piglets and adults, respectively, for the same study area from 1993 until 1995. Our results are in line with the outcomes found in Sáez-Royuela and Tellería (1987) and Fonseca et al. (2004) where age had a positive effect on the litter size. Boitani et al. (1995) however, did not find a relation between litter size and age. Maternal weight as well as maternal age had a high positive influence on the litter size ($p < 0.001$) in this study, which was also found in other research (Fonseca et al. 2011; Rosell et al. 2012). As adults have a significantly higher weight than piglets, they can invest more energy in a higher litter size.

Although there is no clear evidence for a higher percentage of crossbreeding among wild boar in Lower Saxony (Gethöffer et al. 2007), the possibility of hybridization should be taken into account. Overall, hybridization with domestic pigs seems to occur at a very low frequency under natural conditions, but as it is a common practice in farmed stocks, it represents a severe risk for wild populations (Scandura et al. 2011). Studies in Europe indicate inconsistent results: in Greece for instance hybrids within the populations of free-ranging wild boar was described to be quite low (5.0 %) (Koutsogiannouli et al. 2010) whereas in Italy and Corsica, crossbreeding is widespread (Randi 2005).

Both temperature and humidity affects the reproductive physiology and population dynamics in the terrestrial ecosystem (Walther et al. 2002). Climate change causes a higher frequency of mast years in Europe (Övergaard et al. 2007), which has an effect on the population increase in wild boar (Bieber and Ruf 2005; Melis et al. 2006). This might be the reason for the population increase in the whole of Europe. Acorns are high in carbohydrates, which are essential for reproduction and maintenance of

good physical condition (Massei et al. 1996) and can explain our findings about a significantly higher litter size under excellent mast crop conditions. Several studies found a correlation between good mast years and a large litter size as well (Fernández-Llario et al. 1999; Geisser and Reyer 2005).

However, only an excellent mast year of oak and beech can increase the reproduction with a higher litter size, compared to moderate or failure conditions of mast. A moderate condition of oak mast or failure does not have different effects on the litter size. Interactions of oak and beech mast conditions were not significant and therefore not mentioned. Those results indicate that wild boar in Germany live under very good physical conditions, probably because they make use of agricultural food in summer (Keuling et al. 2009). Therefore, they seem to be less dependent on mast years. Even in years of mast crop failure, there is enough food available to bring about a high reproduction. Supplementary feeding by baiting might also have an impact, but in comparison with the food supply by mast years (20 000 kg/year) and agriculture (50 000 kg/year), the amount of maize supplied in winter is minimal (Keuling 2015). Advantageous weather and mast conditions cannot be influenced by hunters, but they can control the artificial supply of maize, which should be handled in a responsible way (Briedermann 2009).

Intensification of agriculture is recognized as one of the most significant human alterations to the global environment (Matson et al. 1997). Although Schley and Roper (2003) argue that mast is preferred above agricultural crop and supplementary food, an additional stable food supply provided by crop fields and supplementary feeding on game feeding places may buffer the negative impact of mast failures and help to keep a high reproduction (Bieber and Ruf 2005; Cellina 2008; Rosell et al. 2012). This statement is supported by a former study from Poland. Here, it was demonstrated that 91 % of the total mass of wild boar food consisted of plants and 71 % of those were cultivated plants (Genov 1981). The most attractive cultivated crops for wild boar were potatoes, oats and maize (Genov 1981). More recent research from Merta et al. (2014) compared diet, carcass weight, and kidney fat of

wild boar in two different areas: a forest-farmland environment and a large compact forest area. Kidney fat and weight of piglets and older animals were higher in the forest-farmland environment. Also Geisser and Reyer (2005) found a higher population density of wild boar with greater maize cultivation as well as Neet (1995) showed in a modelling approach that over a 32 year-period, the expansion of maize culture explained 49.7 % of the variance in the numbers of wild boar.

All those studies support our hypothesis that agricultural dominated areas can favour the litter size.

Groot Bruinderink and Hazebroek (1994) found a relation of weight and mast availability that could indicate an indirect effect of mast via maternal body mass on the litter size. We cannot support the results that different age classes responded in a different way to mast availability, as interactions were not significant. Those different outcomes can be explained by the fact that wild boar in the Netherlands lived under different conditions, namely without access to agricultural fields. Actually, the research of Groot Bruinderink and Hazebroek (1994) also supports our hypothesis because in situations, where wild boar received supplementary feeding, recruitment seemed to be independent of mast availability.

Litter size under moderate and excellent mast conditions of beech differed less from each other than in oak. Probably, beech has a minor effect on reproductive behaviour of wild boar (Cutini et al. 2013). Another possible explanation is the different sample size between moderate and excellent mast crop conditions in beech (Figure 1) and the lower presence of beech compared to oak in the study area. This could make effects less obvious; as a result, the need for high sample sizes and long term studies should be emphasized.

Moisture has an effect on wild boar activity as the ground is easier to root and to pick up scents of food resources (Lemel et al. 2003; Welander 2000). This fits with the fact that more earthworms are found in the diet of wild boar under rainy conditions (Baubet et al. 2003). Even if earthworms are only a small part of the wild boar diet, they supply high nutritional value in terms of protein and vitamin B12 (Keuling, pers.

comm.). Especially the high lysine content is an essential amino acid for the body growth in pigs (Henry et al. 1992). Body growth is related to a higher weight of wild boar and can therefore result in a higher litter size when more earthworms are available for wild boar.

Additionally, in rainy autumns, possibilities of mud wallowing are better for wild boar, which can help to diminish the number of ectoparasites on their fur (Fernández-Llario 2005). This could improve the immune system and physiological condition.

Geisser and Reyer (2005), however, did not find that precipitation had any effect on the population density. Differences might result from the fact that precipitation from another season (April until June) was used as explanatory variable. The results of our study are in line with the outcome of the research of Fernández-Llario and Mateos-Quesada (2005). They found in Western Spain a significant increase in litter size in females, which were older than 2 years, in rainy years compared to dry years. However, since Western Spain is a drier and hotter region than Northern Germany, precipitation could have a much higher impact on the litter size.

A higher temperature in autumn is positively correlated with a higher mast yield of beech in the same year. This could indicate a longer mast period under warmer autumn conditions. Because our study area consists of 55 % agriculture, crops could stay longer in the fields when the temperature in September and October is higher. Wild boar can feed more and for a longer time and thereby gain higher weight to be more successful in reproduction. This result fits with the outcome of Geisser and Reyer (2005), who found a population increase in warmer winters. However, in their research, a favourable temperature was correlated with lower juvenile mortality whereas only better food conditions increased litter size. Nevertheless, the weather conditions seem to have an effect on the litter size of wild boar acting through their nutritional condition.

In our study the HI is seen as an indication for the density of wild boar and therefore has to be regarded with caution. Hunting bag depends on the way of hunting (Keuling et al. 2013b). In the studied population, samples were mainly taken

from wild boar hunted by driven hunts, which are less selective compared to single hunts (Keuling et al. 2013b), and there were no limitations in sex and age (Keuling, pers. comm.). This represents the true population structure in a better way because it reflects the hunting coincidence (Keuling et al. 2013b).

It can be assumed that the age structure and density of the culled wild boar sows are representative for the age structure (Merta et al. 2015) and density (Acevedo et al. 2007) of the studied wild boar population.

We found that the litter size was lower when the HI was high, which could indicate that at high wild boar density, the population is closer to carrying capacity. Therefore, female wild boar produce fewer offspring. Those results are in line with the hypothesis of Massei et al. (1996), who expected that the population density might affect the litter size production. Thus, it seems that wild boar regulate the population themselves by adjusting the number of offspring. Close to carrying capacity, wild boar act like K-strategists, whereas at low population density they act like r-strategists. However, the carrying capacity of wild boar seems to be quite high because there are plenty of food resources. Intra-specific competition for food may not act under those conditions, and the high population size can lead to conflicts in traffic, crop damage, and livestock disease (Frackowiak et al. 2013; Gortázar et al. 2006; Herrero et al. 2006; Lagos et al. 2012; Thurfjell et al. 2015).

Groot Bruinderink and Hazebroek (1994) found that the decrease in wild boar body weight from autumn to winter was greater when population density was high. This could also explain our findings, demonstrating that the population density reduces the litter size acting through the body weight.

In this research, a higher HI in the previous year did not influence the litter size significantly. Those results do not fit with the outcomes documented by Servanty et al. (2011), who showed that female wild boar in a heavily hunted area had a larger litter size than females in a lightly hunted area. Different outcomes might result from varying time length of the study period. In addition, they used hunting pressure in two different periods where the difference

between the weak (1.6 individuals/km²) and high hunting pressure (4.2 individuals/km²) was much higher than in our studied population.

The litter size did not show any significant correlation with the HI of the following hunting year. This could be an indication that hunting is insufficient to limit wild boar population growth (Csányi 1995; Massei et al. 2015; Nores et al. 2008). Another possible explanation for this non-correlation could be that the population density of the next hunting season actually is not higher. Data were collected only until February, so that population decrease by perinatal or postnatal mortality could not be recognised in this study.

Conclusion

To summarize, we can say that the wild boar is a very unique species that is able to adapt to different environmental conditions very well by changing between K -and r-strategies. They profit from our cultivated landscapes with high food supply, where there is still enough food in years of mast failure. Due to global warming, also weather conditions are still more advantageous for wild boar in the future. We expect that wild boar populations will increase further as the body weight per age class already increased in the last 12 years, so that larger litter sizes can be produced by sows in the future.

In this study we found no effect of hunting on the litter size of wild boar. Actually, it seems that hunting is still inappropriate to reduce the population size effectively and, thus, diminish risks of livestock disease, traffic collision and crop damage in the future.

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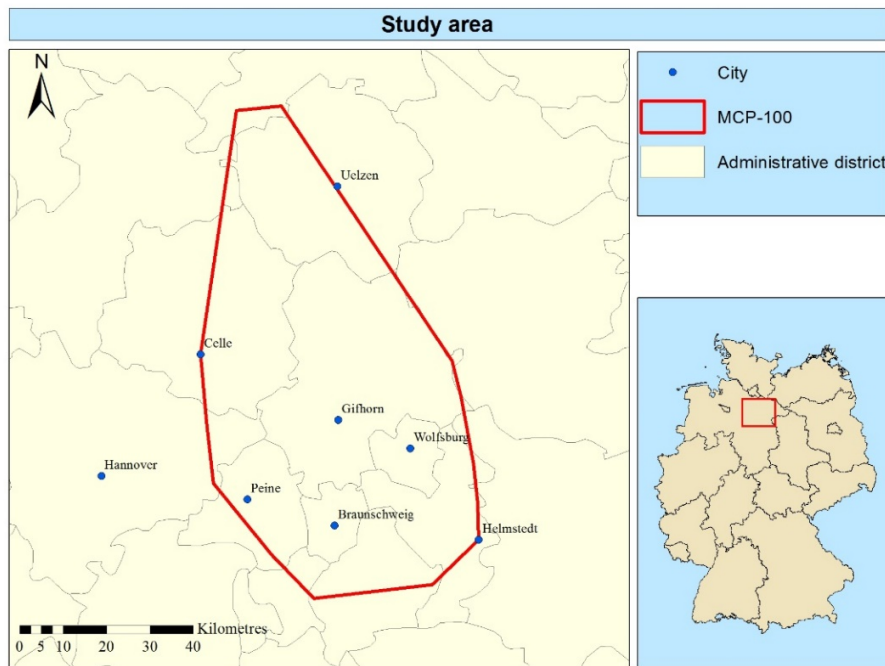
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Appendix

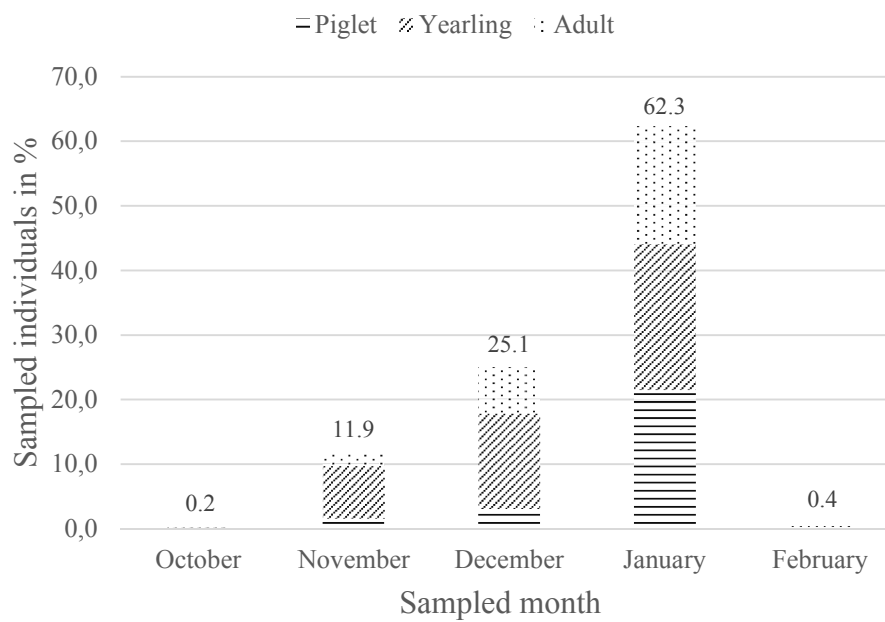
Appendix I:

Map of the study area in Northern Germany, showing a MCP-100 around the sampled locations, some cities as orientation and the administrative districts in the background.



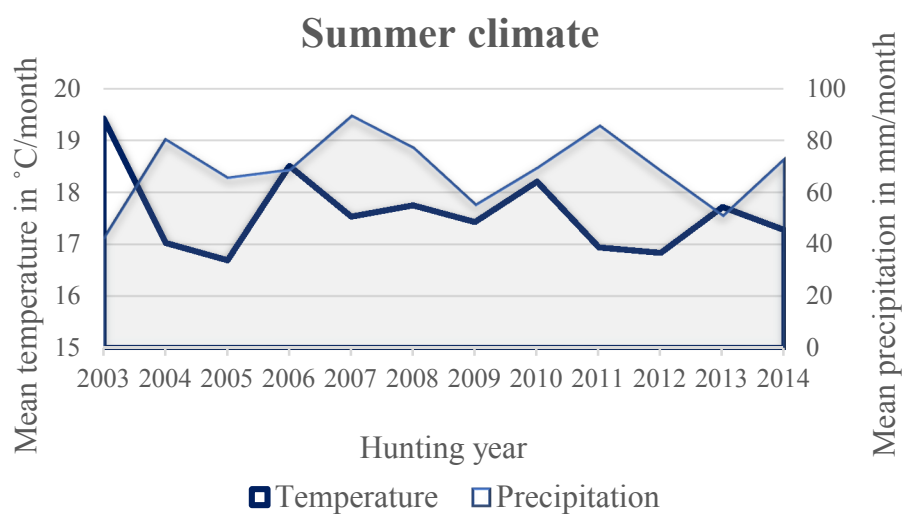
Appendix II:

Distribution of sampled pregnant individuals (in %) per month.



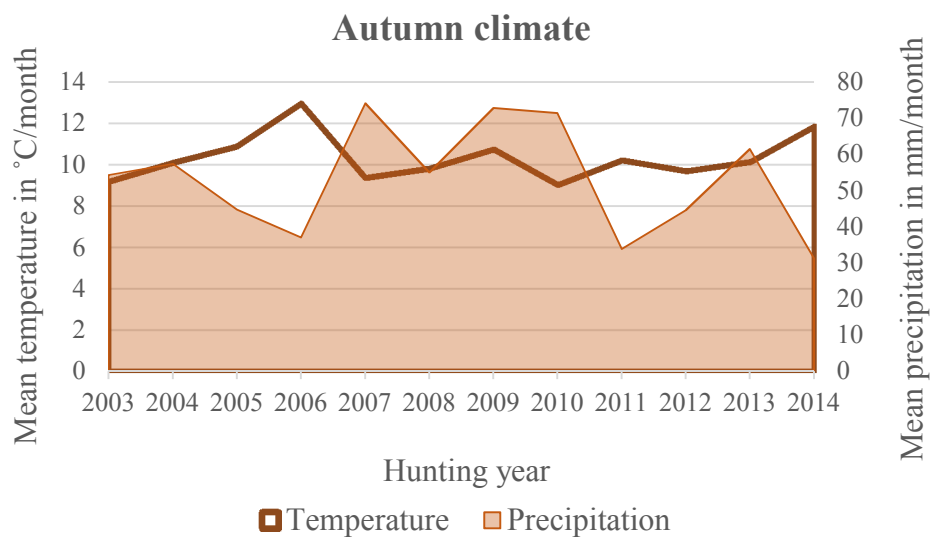
Appendix III:

Mean temperature in °C/month and mean precipitation in mm/month in summer in the study area of the 12 wild boar hunting seasons in Germany.



Appendix IV:

Mean temperature in °C/month and mean precipitation in mm/month in autumn in the study area of the 12 wild boar hunting seasons in Germany.



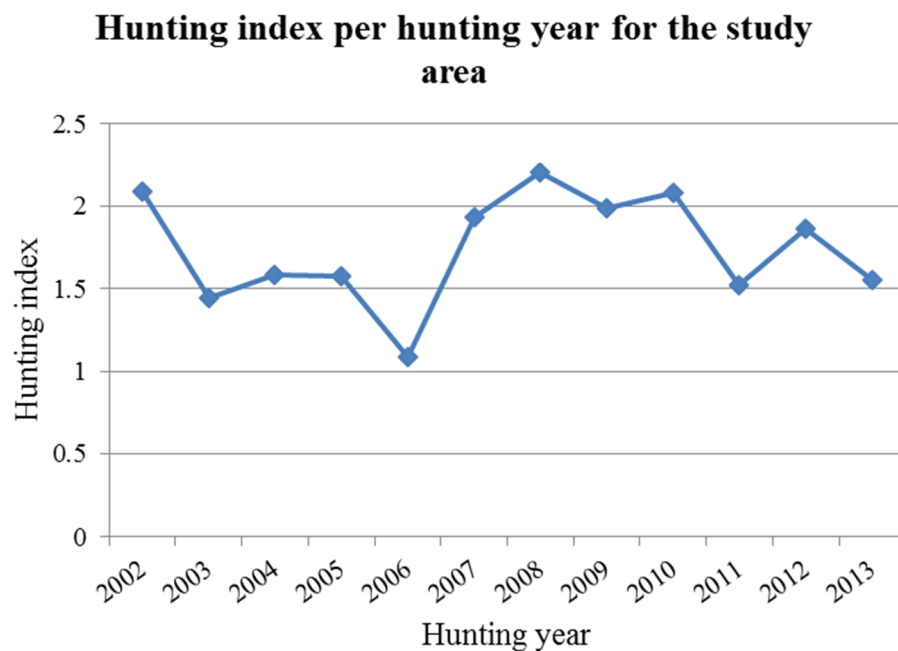
Appendix V:

Mast crop conditions for beech (*Fagus sylvatica*) and oak (*Quercus robur* and *Q. petraea*) per year for Lower Saxony.

Hunting year	Beech	Oak
2003	moderate	excellent
2004	excellent	failure
2005	failure	moderate
2006	moderate	excellent
2007	moderate	moderate
2008	failure	failure
2009	excellent	excellent
2010	failure	failure
2011	moderate	moderate
2012	failure	moderate
2013	failure	moderate
2014	moderate	failure

Appendix VI:

Hunting index (HI=hunting bag/km²) for the study area for the 12 wild boar hunting seasons.



Appendix VII:

Model 1: Weather conditions and age class included as independent variable. Bold model indicates the best fitting model of the round.

Round	Dropped independent variable	Model with included independent variables	AIC
1	None	AgeClass+TemperatureSummer+TemperatureAutumn+PrecipitationSummer+PrecipitationAutumn	1790.613
	AgeClass	TemperatureSummer+TemperatureAutumn+PrecipitationSummer+PrecipitationAutumn	1927.00
	Temperature Summer	AgeClass+ TemperatureAutumn+PrecipitationSummer+PrecipitationAutumn	1790.288
	Temperature Autumn	AgeClass+TemperatureSummer+PrecipitationSummer+PrecipitationAutumn	1793.544
	Precipitation Summer	AgeClass+TemperatureSummer+TemperatureAutumn+PrecipitationAutumn	1788.835
	Precipitation Autumn	AgeClass+TemperatureSummer+TemperatureAutumn+PrecipitationSummer	1790.363
2	None	AgeClass+TemperatureSummer+TemperatureAutumn+PrecipitationAutumn	1788.835
	AgeClass	TemperatureSummer+TemperatureAutumn+PrecipitationAutumn	1925.366
	Temperature Summer	AgeClass+TemperatureAutumn+PrecipitationAutumn	1788.292
	Temperature Autumn	AgeClass+TemperatureSummer+PrecipitationAutumn	1791.614
	Precipitation Autumn	AgeClass+TemperatureSummer+TemperatureAutumn	1788.72
3	None	AgeClass+TemperatureAutumn+PrecipitationAutumn	1788.29
	AgeClass	TemperatureAutumn+PrecipitationAutumn	1925.373
	Temperature Autumn	AgeClass+ PrecipitationAutumn	1794.053
	Precipitation Autumn	AgeClass+ TemperatureAutumn	1791.502

Model 2: Hunting index of the same, previous and following hunting year included as independent variable. Bold model indicates the best fitting model of the round.

Round	Dropped independent variable	Model with included independent variables	AIC
1	none	HISameYear+HIPreviousYear+HIFollowingYear	1673,35
	HISameYear	HIPreviousYear+HIFollowingYear	1675.918
	HIFollwoingYear	HISameYear+HIPreviousYear	1784.885
	HIPreviousYear	HISameYear+HIFollowingYear	1673.687

Model 3: Mast conditions and maternal weight included as independent variable. Bold model indicates the best fitting model of the round.

Round	Dropped independent variable	Model with included independent variables	AIC
1	none	Weight+OakCondition+BeechCondition	1775.329
	Weight	OakCondition+BeechCondition	1915.373
	OakCondition	Weight+BeechCondition	1784.321
	BeechCondition	Weight+OakCondition	1796.374

Manuscript 2

Food conditions influence the effect of population density and maternal weight on foetal sex ratio in wild boar (*Sus scrofa*)

Keywords: *Sus scrofa*, foetal sex ratio, Trivers and Willard hypothesis, population density, nutritional condition

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Abstract

The Trivers and Willard hypothesis (TWH) is the most tested model in polygynous mammals. It assumes that a decline in maternal conditions in general leads to the fact that adult females produce a lower ratio of males to females because weak female offspring will have a higher probability of future reproduction than weak male offspring. Support for this hypothesis in mammals has been equivocal. As the original theory applies to species with only one offspring per litter, it is unclear whether the TWH can be applied to polytocous species because a trade-off between the number and the sex of the offspring can alter the appropriate pattern of the investment in male and female offspring and, therefore, the application of the THW. Also other factors like population density and litter size may contribute to sex-ratio variation within the species of wild boar. As we know that wild boar respond well to food availability and it is expected that they adjust the sex ratio to the less expensive sex under resource limitation, food availability was included as interaction to all parameters (maternal weight, population density and litter size).

During 12 hunting seasons of data collection (2003-2014), the sex of foetuses was determined and the number of foetuses was counted in 205 litters of hunted wild boar. Dressed weight of female wild boar was also taken. Data on the amounts of crops of acorn and the hunting index (HI=hunting bag/km²), which was used as an indication for population density, were available. Binomial distributed generalized linear models were analysed in R.

The study shows that the TWH is not applicable in wild boar since maternal body weight and foetal sex ratio did not correlate. However, in years of mast crop failure, the TWH is applicable in wild boar as the variation of mother's weight is higher than in excellent mast crop years so that the effect of body weight on reproduction is higher. The fact that multiple paternity occurs in wild boar makes it to a less suitable candidate for the TWH as it is doubtful if strong male offspring really have better chances of successful reproduction than strong female offspring. In addition, more males were produced by sows under higher populated conditions in years of mast crop failure. Only a statistical trend ($p < 0.1$) was found for the fact that in years of mast failure, more males were produced by sows with higher litter size. There seems to be a benefit in producing males under limited food conditions since wild boar males are more likely to disperse further than females and are less likely to compete for local resources with the mother.

Introduction

Sex ratio theory has become a rich and diverse area of evolutionary ecology (Hardy 2002). Extraordinary sex ratios are mainly noted in invertebrates and seem to be the exception in birds and mammals. This is why sex ratio variation in vertebrates is still poorly understood (Hardy 2002) and remarkably resistant to adaptive explanations (Krackow 1995). The most tested model in polygynous mammals is the Trivers and Willard hypothesis (TWH) (Trivers and Willard 1973). It assumes that a decline in maternal conditions in general leads to the fact that adult females produce a lower ratio of males to females because weak female offspring will have a higher probability of future reproduction than weak male offspring (Trivers and Willard 1973). Assumptions that should be fulfilled for the application of the TWH are: 1) Offspring quality is correlated with maternal quality. 2) Offspring quality will persist into adulthood. 3) Adult quality affects mating success of one sex more than it does for the other. Empirical support for this hypothesis in mammals has been equivocal (Blanchard et al. 2004; Veeroja et al. 2010) and the original theory applies to species in which the litter size is one (Frank 1990; Hardy 1997).

The wild boar (*Sus scrofa*) is a large, polygynous and sexually dimorphic mammal, which makes it a good candidate for the TWH. However, it is a polytocous mammal (with a mean litter size of 6.6 in these climes) (unpublished, Frauendorf et al.), which is higher than in other similar-sized ungulates (Carranza 1996). Previous tests showed inconsistent results concerning the TWH in species with several offspring per litter (Servanty et al. 2007). Even though different studies documented a foetal sex ratio bias for males as well as for females in wild boar (Keuling et al. 2013a), the TWH was studied only a few times in wild boar and in none of them the TWH could be supported (Fernández-Llario et al. 1999; Servanty et al. 2007). The mother's trade-off between size, number and sex of the offspring (Servanty et al. 2007)

complicates the prediction about the link of maternal condition and sex ratio variation (Fernández-Llario et al. 1999), which is similar to birds as they also have a large litter size (West and Sheldon 2002).

In addition, probably, not only maternal condition influences the foetal sex ratio because long-term studies came to the conclusion that sex ratio variation is subject to multiple environmental influences (Hardy 2002; Kruuk et al. 1999; Servanty et al. 2007; Toro et al. 2006). Explaining such variation across taxa in line of the sex ratio theory remains a major challenge because of multiple factors influencing sex ratio evolution and complex life histories (West and Sheldon 2002).

We expect that maternal condition (Sheldon and West 2004), litter size (Frank 1990; Servanty et al. 2007) and density (Kruuk et al. 1999) might influence the sex ratio in wild boar. Kruuk et al. (1999) found in red deer (*Cervus elaphus*) that the proportion of males born each year declined with increasing population density, which is associated with nutritious stress. As it is commonly known that wild boar respond well to good food availability in terms of mast (Jędrzejewska et al. 1994), and that oak is of main importance (Cutini et al. 2013), we will look at those different parameters considering different oak mast conditions. We expect that under resource limitation (in years of mast failure) wild boar will adjust sex ratio towards the less expensive sex (Myers 1978), thus female offspring, as they have a lower foetal weight than male offspring (Fernández-Llario et al. 1999).

This research contributes efficiently to studies on foetal sex ratios by 1) testing the TWH with focus on a highly polytocous species and 2) examining maternal physiology (maternal body weight), population density and litter size simultaneously (in one study population) that account for between-year variation in foetal sex ratio under different food conditions.

Material and methods

Study area

The study area is situated in the east of the federal state of Lower Saxony (Northern Germany) (52.36°N, 10.35°E) (Appendix I), where the altitude ranges from 60 to 130 m ASL (Gethöffer 2005). Based on data from the German Meteorological Service, the average annual temperature from 2000 until 2010 was 9.7°C (DWD 2015). The average annual precipitation per year was 807 mm from 2000 until 2010 (DWD 2015).

The size of the study area is about 4885 km². Agriculture and forest comprise 54.6 % and 27.6 % of the area, respectively. The remaining 17.8 % are buildings and infrastructure (NLS 2007). The agricultural land is characterized by 43.3 % grain (oat, barley, rye and wheat), 16.0 % grassland, 11.7 % sugar beets, 6.6 % oilseed rape, 6.0 % maize, and 5.8 % potatoes. The remaining 10.6 % consists of heathland, fallow land and horticultural products (NLS 2007).

The forest consists of 18.9 % oak (*Quercus* spp.), 11.2 % beech (*Fagus sylvatica*), 7.6 % other deciduous trees (e.g. *Betula* spp, *Acer* spp., *Tilia* spp., *Sorbus aucuparia*), 45.5 % pine (*Pinus sylvestris* and *P. strobus*) and 16.6 % other coniferous trees (*Larix* spp., *Pseudotsuga menziesii*, *Picea abies*).

Supplementary feeding is not allowed in the study area. However, baiting in the winter with the help of maize is a necessary hunting strategy (Keuling 2015). Approximately not more than 1000 kg supplemental food per 100 ha is supplied every year for baiting purposes (similar to Keuling et al. 2008) whereof about 200 kg are actually eaten by wild boar. This is minimal in comparison with the food supply by mast years and agriculture, which is about 20 000 kg and 50 000 kg per year, respectively (Keuling 2015). Nevertheless, the possible impact of supplementary feeding should not be ignored.

Data collection

In total 3011 reproductive tracts (uteri and ovaries) were collected from free-ranging hunted wild boar sows in the study area in the

hunting years 2003 until 2014. From 455 pregnant sows, 205 were used in analysis and sampled from October until February, where 11.9 %, 25.1 % and 62.3 % were collected in November, December and January, respectively. The remaining part (0.7 %) was sampled in October and February. The hunting season 2009, for instance, starts in October 2009 and ends in February 2010. Each individual sampled is associated with a locality and thus with one of the nine administrative districts in the study area (Appendix I). Dressed weight (without intestines) of female wild boar was taken. During dissection in the laboratory, the number of fetuses was counted. Sex could be determined without ambiguity macroscopically from a foetal age of 30 days from the moment of conception based on the presence of a vulva in females and a penis in males. Sample size, distribution of the age classes as well as total sampled fetuses differed per hunting year (Appendix II).

Data on amounts of acorn crop size were provided by The Forestry Seeds Advisory Board (FSB) at Oerrel/Lower Saxony, Germany. The yield values of oak (*Quercus robur* and *Q. petraea*) per year (from 1 July until 30 June) for Lower Saxony were transformed in three classes indicating the condition of the mast crop. Excellent mast crop demonstrates > 70 % of the maximum possible harvest, moderate mast crop condition indicates 10-69 % of the maximum possible harvest, and failure is defined as < 10 % of the maximum possible harvest (Keuling et al. 2013b). Mast crop conditions were excellent in 2003, 2006 and 2009 whereas 2004, 2008, 2010 and 2014 were years of failure in acorns.

Each sampled individual is also associated with a hunting index (HI) of the administrative district where it was collected. The HI is defined as the hunting bag per km² and as data were mainly collected with the same hunting technique, we expect that effort was the same in each year. From 2003 until 2005, the HI increased but decreased in 2006 to a value of about 1.2. In the next years, the HI increased and decreased again in 2011 to a value of about 1.5, followed by a peak of 2.0 in 2012 and a low level in 2013 and 2014 (Appendix III).

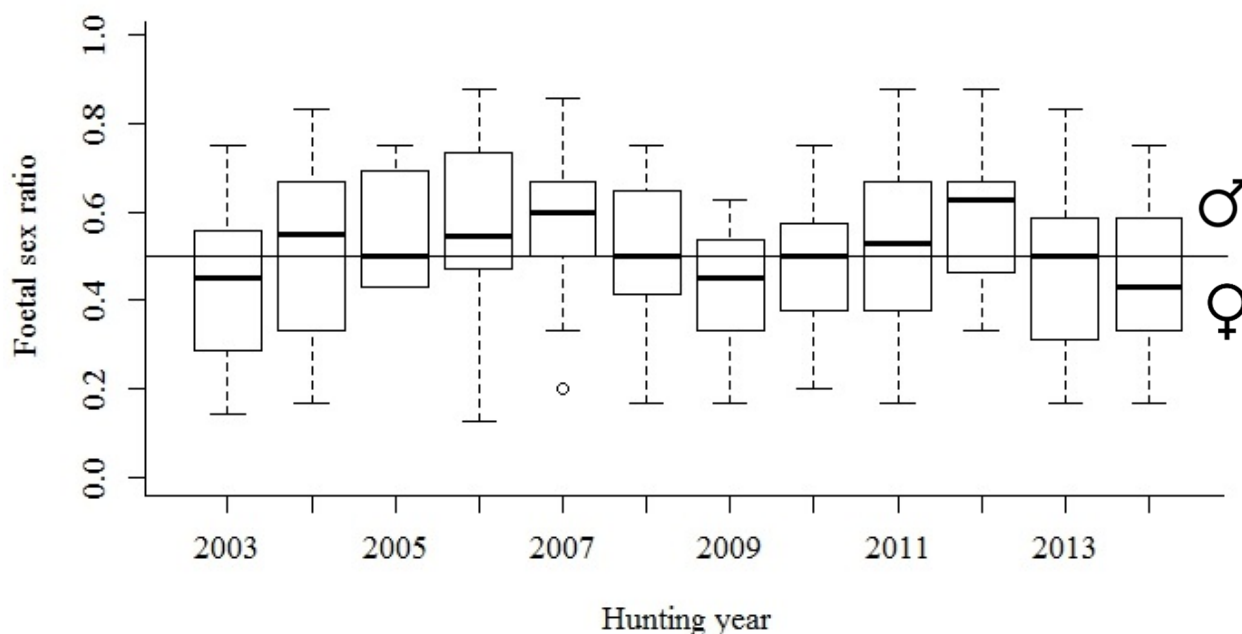


Figure 1: Foetal sex ratio per hunting year. 0.5 indicating no bias in foetal sex ratio. Sex ratio >0.5 means male bias and sex ratio <0.5 shows female bias. The horizontal line shows sex ratio equality. Box and whisker plots show median (horizontal line within box), 25% (interquartile range 1) and 75% (interquartile range 3) percentiles (box) and the range (whiskers), which includes observations up to 1.5 times the interquartile ranges. Circles indicate statistical outliers (observations outside 1.5 times the interquartile ranges). Wild boar foetal sex ratio over the 12 hunting years did not differ significantly from equality.

To be able to analyse the effect of wild boar density on foetal sex ratio, estimations of the abundance of wild boar in the study area are necessary. Unfortunately, the expense and effort that is required to estimate true population densities are prohibitive (Acevedo et al., 2007). Data required by trap cam method (Rowcliffe et al. 2008) are only partly available for the study area so that the HI has to be taken as an indication for the population density of wild boar in the study area. The HI is regarded as an appropriate measurement to obtain valid results about the wild boar density (Acevedo et al. 2007).

Statistical analysis

For all tests, significance was assumed when $p < 0.05$. All analysis were performed using R software version 3.1.2 (R Core Team 2014).

The foetal sex ratio is best expressed in proportion (sex ratio = males / (males + females)) where it is treated as grouped binary data (Hardy, 2002). A sex ratio of 0.5 means an equal sex ratio of 1:1. A sex ratio >0.5 indicates a male biased sex ratio whereas a sex ratio <0.5 indicates a female biased ratio. Deviations are

expected to conform a binomial distribution (Zuur et al., 2010; Hardy, 2002). Based on our hypothesis, three logistic regressions (maternal weight, HI and litter size as independent variable) were performed to find out which variables explain changes in foetal sex ratio. In each model the mast crop condition was included as interaction because we expected different foetal adjustment in different mast conditions. Models were checked for Pearson correlations and validated by checking Cook's distance and temporal autocorrelation. Models were corrected for underdispersion because pearson dispersion statistic was under 1 (Hilbe 2014).

Results

The mean foetal sex ratio during the whole study period was 0.5 (± 0.18 SD). This ratio shows foetal sex equality either by considering total number of male and female foetuses (chi-square test: $\chi^2 = 0.003$, $df = 1$, $p = 0.955$) or by comparing the number of male and female foetuses per litter (Wilcoxon test: $z = -0.219$, $n = 205$, $p = 0.826$). However, foetal sex ratio of wild boar differed in the 12 years (Figure 1).

Weight only showed a positive trend to a higher male production (GZLM, $t=1.868$, $n=205$, $p=0.0632$). Excellent mast crop years however increased the production of male fetuses significantly (GZLM, $t=2.098$, $n=205$, $p<0.05$) compared to years of mast failure. Additionally, more males were produced by heavier wild boar sows in a year of mast crop failure of oak whereas in years of excellent mast crop conditions, heavier animals produced more female offspring (GZLM, excellent mast

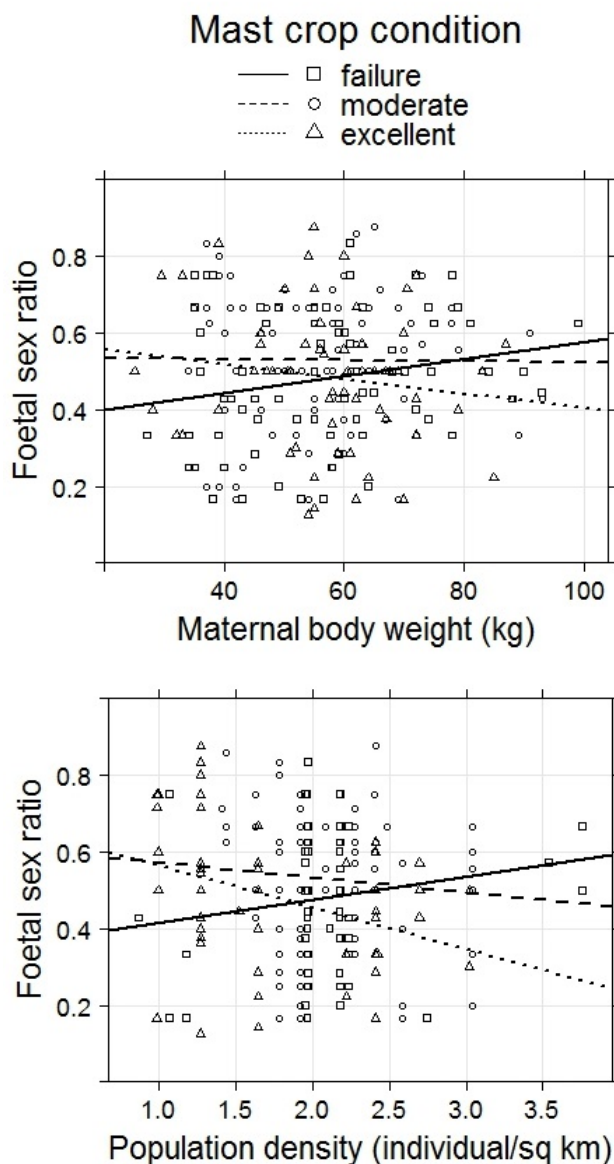


Figure 2: Foetal sex ratio in relation to maternal body weight (in kg) and population density (individual/km²), respectively, per mast crop condition of oak (*Quercus robur* and *Q. petraea*) of 12 wild boar hunting years. 0.5 indicating no bias in foetal sex ratio. Sex ratio > 0.5 means male bias and foetal sex ratio < 0.5 shows female bias. Dots are observed values per mast crop condition. Fitted values (solid, dashed and dotted line) are obtained by a binomial glm.

crop of oak in interaction with weight: $t=-2.213$, $n=59$, $p<0.05$) (Figure 2).

No significant difference in foetal sex ratio was found between moderate mast crop conditions in interaction with maternal weight and years of failure (GZLM, moderate mast condition of oak in interaction with weight: $t=-1.336$, $n=76$, $p=0.1832$).

Also the second model indicated a higher male offspring production in good mast years compared to years of failure (GZLM, $t=2.356$, $n=205$, $p<0.05$). Population density had no effect on foetal sex adjustment in wild boar (GZLM, $t=1.125$, $n=205$, $p=0.26192$). However, the interaction of mast condition and population density was highly significant indicating that sows living under higher population density produced more females in excellent mast crop years whereas they produced more males in years of mast failure (GZLM, excellent mast crop condition in interaction with density: $t=-2.632$, $n=59$, $p<0.01$) (Figure 2). No significant difference in foetal sex ratio was found between moderate mast crop conditions in interaction with population density and years of failure (GZLM, moderate mast condition in interaction with density: $t=-1.157$, $n=76$, $p=0.24862$).

The litter size had no significant influence on the foetal sex ratio (GZLM, $t=0.811$, $n=205$, $p=0.4185$). In interaction with mast crop condition however, it showed an influencing trend on foetal sex ratio. A higher litter size resulted in male biased and female biased sex ratio in failure and excellent mast crop conditions, respectively (GZLM, excellent mast condition in interaction with litter size: $t=-1.725$, $n=59$, $p=0.0861$). Sex ratio in moderate mast crop years did not differ significantly from failure years (GZLM, moderate mast condition in interaction with litter size: $t=0.127$, $n=76$, $p=0.8993$).

Discussion

Trivers and Willard hypothesis

There is abundant evidence that among invertebrates, adults manipulate the sex ratio of their offspring to maximize their own fitness (Clutton-Brock and Iason 1986) and it is

expected that mammals behave similarly according to the TWH (Hardy 2002). The studied wild boar population only showed a trend of a correlation between maternal weight and foetal sex ratio so that we cannot support the TWH. Male foetuses were on average 6 % heavier than female foetuses in the studied population, which is similar to other wild boar populations in Europe (Fernández-Llario et al. 1999). Thus, a possible explanation for the non-correlation of maternal condition and foetal sex ratio could be that mothers transfer a greater portion of their body reserves to sons than to daughters (Clutton-Brock et al. 1981), but do not adjust the sex ratio. Also other studies on wild boar (Fernández-Llario et al. 1999; Servanty et al. 2007), Iberian pig (Toro et al. 2006) and domestic pig (Mendl et al. 1995) could not support the outcome of the TWH hypothesis by looking at maternal weight in relation to sex ratio in the foetal stage or at birth. However, Sheldon and West (2004) used a meta-analysis to review published studies on different ungulates (among others fallow deer (*Dama dama*), roe deer (*Capreolus capreolus*), red deer (*Cervus elaphus*), moose (*Alces alces*), reindeer (*Rangifer tarandus*), bighorn sheep (*Ovis canadensis*), bison (*Bison bison*), pronghorn (*Antilocapra americana*), wild boar (*Sus scrofa*), mountain goat (*Oreamnos americanus*)) and found a positive correlation between maternal condition and sex ratio. Different results concerning the TWH might be based on different measures of maternal quality (age, weight or social rank) (Blanchard et al. 2004).

However, in years of mast crop failure, the TWH is applicable in wild boar. Sows in better physiological condition (higher body weight) produce more males when food availability is low. Only stronger sows are able to produce strong male offspring that will be successful in future reproduction. Variation of maternal body weight is higher in failure mast years (+/- 16.3) than in excellent years (+/- 13.6). This explains the fact the TWH is only applicable for wild boar in those years of mast failure when the effect of weight on the reproduction is higher. In addition, males are the larger sex and suffer higher mortality rates before the end of parental

care (Toro et al. 2006) due to higher sensibility to harsh environmental conditions (Clutton-Brock and Iason 1986), so that in years of mast failure, only sows in better physiological condition are able to produce sons that are capable to survive the critical phase.

The results are supported by Wright et al. (1988), who found out that food restricted female house mice which were socially dominant produced more males whereas subordinate mice produced more female offspring. This also supports the hypothesis that male offspring are probably more sensible to starvation so that only dominant mothers are able to produce successfully male offspring under nutritious stressed situation (Krackow 1995).

Sheldon and West (2004) argue that behavioural measure of dominance supported the TWH more strongly than weight or age. Thus, the TWH prediction then may only apply to species where female access to resources directly responds to social rank (Blanchard et al. 2004). Under good food conditions, feeding sites are not defended by dominant wild boar (Dardaillon 1988; Graves 1984) so that we can assume that there is no correlation between maternal weight and social rank in the studied population in good mast years and therefore the TWH does not apply. Probably, high food availability might be a reason for the non-application of the TWH.

In addition, as Kruuk et al. (1999) argued that all support for the TWH has been found in ungulate populations below carrying capacity, it can be supposed that the studied wild boar populations did not reach carrying capacity yet, at least in years of high food supply.

The third assumption of the TWH expects that in polygynous species males compete more intensively for access to mates than females do (Trivers and Willard 1973). Thus, an advantage in the condition of male offspring can increase the future reproductive success more than in female offspring (Trivers and Willard 1973). However, there is evidence for the regular occurrence (22.8 %) of multiple paternity in wild boar in Germany (unpublished, Müller et al.), which may lead to the insight that wild boar are not perfect candidates to fulfill the

assumptions of the TWH.

Additional parameters affecting foetal sex ratio

With increasing population density, wild boar females produced more male offspring under food restricted conditions whereas at food abundance more female fetuses were found.

The results under abundant food resources are in line with outcomes in red deer populations in Kruuk et al. (1999) where more female offspring are produced at higher population density. They argue that due to nutritious stress-situations (at high density), more females were born which is less expensive in production. We assume that due to a high food supply mechanisms work different in wild boar.

The local resource competition (LRC) hypothesis (Silk 1983) could explain our results under food restricted conditions. Females under poor condition (in years of mast failure) and in years of high population density would benefit from producing males, as wild boar males are more likely to disperse further than females (Keuling et al. 2010) and, therefore, adult sons are less likely than adult daughters to compete with their mother for local resources (Blanchard et al. 2004). If the population would become extinct (due to food shortage), the genes in dispersed male offspring would survive at least.

In excellent mast years, there is no nutritious stress for the animals and more female offspring (that disperse less than males) can be produced even if population density is high.

Thus, we can hypothesize that carrying capacity in years of high food supply is not reached yet in the studied population as already mentioned above. Density dependent stress parameters do not trigger mechanism in wild boar for foetal sex bias, but mainly food limitation may act here.

Results only showed a trend of a relationship ($p < 0.1$) between foetal sex ratio and litter size which was also described by other researchers (Frank 1990). In years of mast failure, more male fetuses were found at higher litter size whereas under excellent mast conditions, more females were produced at higher litter size.

Thus, in years of high food supply, when sows produce large litter size, there are less

resources for each individual offspring, so that it is of poorer condition and, therefore, it is advantageous for successful future reproduction to invest in female offspring (Servanty et al. 2007).

It might be better in years of failure to produce more males when litter size is high as based on the LCR hypothesis (mentioned above), males are more likely to disperse and, therefore, will compete less for the low food resources with the rest of the family group.

Often the body size or social rank is used as major determinant of reproductive success. However, if it would be instead inherited behavioural characteristics, and large litters are an indication for high competitive ability of the mother, we would expect a male biased litter in years of food restriction (Wright et al. 1988), what we found in this study.

Fernández-Llario et al. (1999) did not find any correlation of litter size and foetal sex ratio in wild boar. Maybe because the observed range of litter sizes was lower (1-7) than in our studied population (3-11).

Conclusion

According to this study, one can conclude that the TWH is only applicable in wild boar under harsher environmental conditions like in years of mast crop failure and, therefore, when population is closer to carrying capacity. Then, only heavier animals can invest in male offspring, which is the more expensive sex because male fetuses are on average heavier than female fetuses. The fact that the wild boar is a polytocous species and foetal sex ratio is triggered by several parameters (weight, food availability, density and litter size) makes the prediction of the TWH more complicated. In addition, since multiple paternity occurs in wild boar, it is doubtful if this species is an ideal candidate for the TWH. All parameters interact with food condition and confirm that resource limitation is the main parameter influencing mechanism in sex ratio variation. We also want to emphasize that there is importance of litter size adjustment in species with several offspring per litter that may be more relevant than adjusting foetal sex ratio.

Acknowledgement

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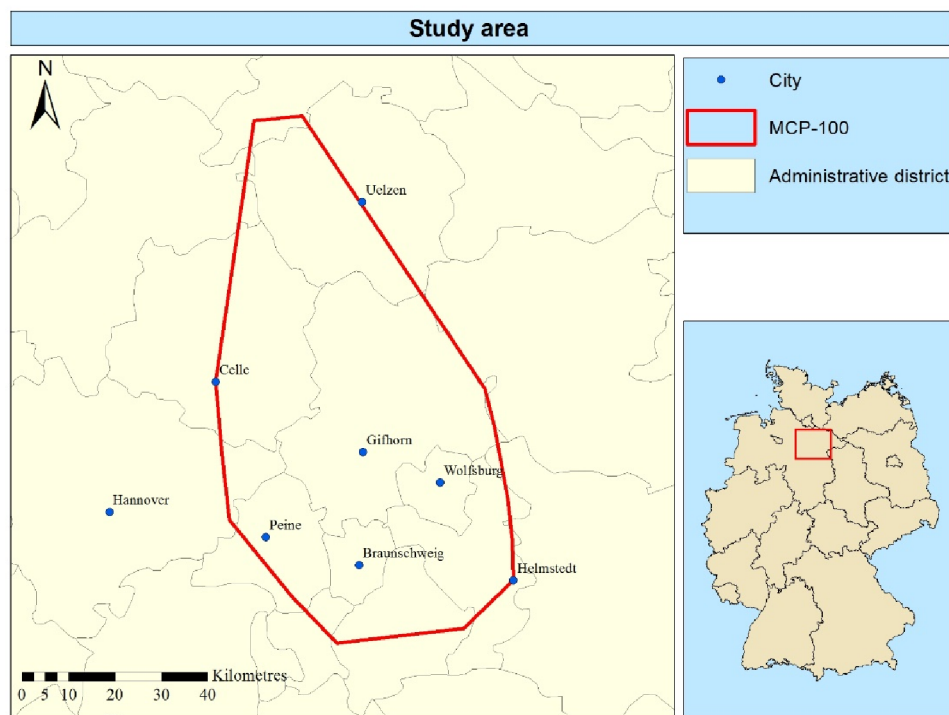
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Appendix

Appendix I:

Map of the study area in Northern Germany, showing a MCP-100 around the sampled locations, some cities as orientation and the administrative districts in the background.



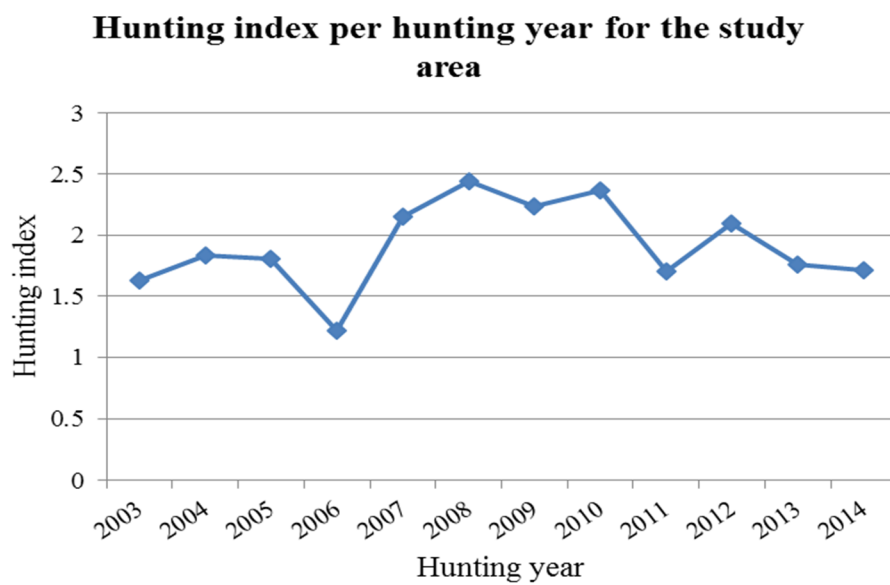
Appendix II:

Sampled individuals (per age class in %) and total sampled fetuses per hunting year. Age was identified by tooth determination.

Hunting year	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Sampled individual sows	22	14	7	23	15	15	14	14	20	19	15	27
Piglets (≤12 months)	18.2	28.6	14.3	17.4	40.0	20.0	12.5	28.6	18.2	15.8	40.0	39.3
Sampled individuals per age class in %												
Yearlings (13-24 months)	45.4	35.7	57.1	47.8	40.0	40.0	56.3	50.0	50.0	63.1	46.7	35.7
Adults (>24 months)	36.4	35.7	28.6	34.8	20.0	40.0	31.2	21.4	31.8	21.1	13.3	25.0
Total sampled fetuses	153	101	44	170	96	100	95	101	141	123	90	159

Appendix III:

Hunting index (HI=hunting bag/km²) for the study area for the 12 wild boar hunting seasons.



Manuscript 3

The effect of environmental and physiological factors as well as litter size and hunting on the timing of birth in wild boar (*Sus scrofa*) in Germany

Keywords: *Sus scrofa*, birth date, synchronous reproduction, maternal condition, body weight, age, weather, hunting, population density, nutritional condition, food availability, mast years, agriculture, conception date, generalized linear mixed model (GLMM)

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Abstract

The wild boar (*Sus scrofa*) is a popular game species in Europe. As problems regarding the overlap of hunting season and breeding season on wild boar are very common in almost all European countries, it is important to know which factors influence the timing of birth and what kind of changes we can expect in the future so that the hunting season can be adapted concerning animal welfare. Therefore, the purpose of this study was to analyse how physiological and environmental factors as well as hunting and litter size influence the timing of birth of wild boar populations in Northern Germany.

During 12 years of data collection (2003-2014), 444 reproductive tracts of hunted pregnant wild boar were sampled (from October to February). During dissection of the reproductive tract, the size of foetuses was measured so that birth date could be calculated. The dressed weight of female wild boar was taken, and the age was examined by tooth determination. Additionally, the age of 1682 culled piglets could be examined based on tooth determination so that their birth date could be identified throughout the year, either. To find parameters influencing timing of birth, meteorological data, data on the amounts of acorn crop size and hunting index (HI=hunting bag/km²) were available. General and generalized linear mixed models depending on the research question were analysed in R.

Wild boar showed a low degree of synchronous breeding compared to other large ungulates. Probably, they live under good nutritious conditions (among others due to agriculture intensification) so that forage conditions are less restricted to a certain period of the year. Peaks of birth occurred in spring (March), and only 4 % of the piglets were born during summer. The birth date occurred earlier when the sows had a higher age, larger body weight and larger litter size, and when the temperature in autumn was higher as well as under better mast crop conditions. Thus, wild boar adapt conception time on several environmental and physiological conditions. Parturition will start earlier in the future so that hunting season might be shortened or advanced. Hunting and the population density had no effect on timing of birth. The probability that wild boar are born in summer was higher when weather in February was cold and wet. Probably, new-born piglets are sensitive to harsh weather conditions at time of birth and when sows lose their litter, they start reproducing a second time in the same year.

Introduction

The wild boar (*Sus scrofa*) is a popular game species (Uzal and Nores 2004) that has increased in the whole of Europe in the last decades (e.g. Keuling et al. 2013a; Lombardini et al. 2014; Massei et al. 2015).

Factors determining the various aspects of the reproductive biology are still the main area of research (Fernández-Llario et al. 2003) and problems regarding the overlap of hunting and breeding season on wild boar are very common in almost all European countries (Fonseca et al. 2001). Even though there are limited hunting seasons for adult females and males, and sows of offspring are not allowed to be culled, the prevention of an overlap of hunting and breeding season can ensure the welfare for the animals.

There are contradictory results concerning seasonal synchrony and the number of reproductive periods in wild boar (Santos et al. 2006). Frequency and timing of birth peaks depend on the country or geographic area (Briedermann 2009). Nevertheless, wild boar offspring are mostly born in the period December until May (Aumaitre et al. 1982; Fernández-Llario and Mateos-Quesada 1998; Fonseca et al. 2011; Gethöffer et al. 2007). Thus, it seems that wild boar show an asynchronous reproductive behaviour with a wide variability in month of birth, which may be influenced by climatic conditions (Fernández-Llario and Carranza 2000). Temperature and precipitation at time of conception and during pregnancy might influence the birth date (Forchhammer et al. 2001). Also higher maternal condition (Briedermann 2009), better food availability in terms of mast condition (Maillard and Fournier 2004; Pépin et al. 1987) as well as a larger hunting bag (Gamelon et al. 2011) led to earlier parturition in wild boar.

However, no research has focused on all those parameters in one study area.

Additionally, high juvenile mortality was noticed during or after a harsh winter (Jędrzejewski et al. 1992). Piglets are susceptible to unfavourable weather conditions

like low temperature (Geisser and Reyer 2005) and frost at time of birth and after birth in winter and spring (Howells and Edwards-Jones 1997). Furthermore, high spring precipitation was related to perinatal mortality in ibex (Escós and Alados 1991) and domestic pig (Akos and Bilkei 2004). Compared to other ungulates, wild boar are less resilient to such environmental conditions (Okarma et al. 1995). When the sow lost offspring, she started breeding again in order to bear a second litter (Briedermann 2009). Also the occurrence of birth in summer months is documented discordantly (Durio et al. 1995; Lombardini et al. 2014; Maillard and Fournier 2004; Moretti 1995), and no studies have examined possible factors that may lead to adjustment of the reproductive behaviour in wild boar.

This research contributes to studies on birth timing behaviour in wild boar by 1) testing if birth dates in wild boar is evenly distributed throughout the year 2) examining which factors influence timing of birth in the winter/spring period and 3) finding out if summer birth occurs and if it is affected by winter weather conditions.

Material and Method

Study area

The study area is situated in the east of the federal state of Lower Saxony (Northern Germany) (52.36°N, 10.35°E) and has a size of about 4885 km² (Appendix I). Altitude ranges from 60 to 130 m ASL (Gethöffer 2005). Based on data from the German Meteorological Service, the average annual temperature from 2000 until 2010 was 9.7°C (DWD 2015). The average annual precipitation per year was 807 mm from 2000 until 2010 (DWD 2015).

Agriculture and forest comprise 54.6 % and 27.6 % of the area, respectively. The remaining 17.8 % are buildings and infrastructure (NLS 2007). The agricultural land is characterized by 43.3 % grain (oat, barley, rye and wheat), 16.0 % grassland, 11.7 % sugar beets, 6.6 % oilseed rape, 6.0 % maize, and 5.8 % potatoes. The remaining 10.6 % consists of heathland, fallow land and horticultural products (NLS 2007).

The forest consists of 18.9 % oak (*Quercus* spp.), 11.2 % beech (*Fagus sylvatica*), 7.6 % other deciduous trees (e.g. *Betula* spp, *Acer* spp., *Tilia* spp., *Sorbus aucuparia*), 45.5 % pine (*Pinus sylvestris* and *P. strobus*) and 16.6 % other coniferous trees (*Larix* spp., *Pseudotsuga menziesii*, *Picea abies*).

Supplementary feeding is not allowed in the study area. However, baiting in the winter with the help of maize is a necessary hunting strategy (Keuling 2015). Approximately not more than 1000 kg supplemental food per 100 ha is supplied every year for baiting purposes (similar to Keuling et al. 2008) whereof about 200 kg are actually eaten by wild boar. This is minimal in comparison with the food supply by mast years and agriculture, which is about 20 000 kg and 50 000 kg per year, respectively (Keuling 2015). Nevertheless, the possible impact of supplementary feeding should not be ignored.

Data collection

In total 3011 reproductive tracts (uteri and ovaries) were collected from free-ranging hunted wild boar sows in the study area in the hunting years 2003 until 2014. Hunting season 2009, for instance, starts in October 2009 and ends in February 2010. Each individual sampled is associated with a locality and thus with one of the nine administrative districts (Appendix I). Dressed weight (without digestive system, heart, lungs, liver, reproductive tract and blood) of female wild boar was taken. The age of the wild boar sows was identified by tooth determination (Gethöffer et al. 2007). The lower jaw of the sampled individuals was examined, and the tooth eruption, replacement, and dental abrasion were assessed. Based on this age identification, wild boar sows were pooled in three classes (piglets: ≤ 12 months, yearlings: ≥ 13 -24 months, adults: ≥ 25 month). During dissection in the laboratory, the crown-rump length of the foetuses was measured.

As weather conditions at time of birth and time of conception might influence the timing of birth (see introduction), meteorological data (temperature and precipitation per month) were supplied by local meteorological services

(DWD 2015). The mean temperature and precipitation of the two seasons, autumn (September-November) and winter (December-February) during data collection were calculated. Each sampled individual is associated with an average temperature and precipitation per month in autumn and winter, which depends on the hunting year and administrative district where the individual sample was collected. Each administrative district was associated with an appropriate weather station close by.

In 2006, 2011 and 2014, the mean autumn precipitation was low whereas it was high in 2007, 2009 and 2010 (Appendix II). The years 2003, 2007 and 2010 had a relatively cold autumn whereas in 2006 the mean autumn temperature was quite high (Appendix II).

Winter climate data indicate dry winters in 2005, 2008 and 2013 whereas the winters of 2003, 2006 and 2011 were relatively wet (Appendix III). The winters were relatively warm in 2006 and 2013 but cold in 2005, 2009 and 2012 (Appendix III).

Data on amounts of acorn crop size were provided by The Forestry Seeds Advisory Board (FSB) at Oerrel/Lower Saxony, Germany. Oak was used as mast seeding tree as we expected beechnut to have a minor effect on reproductive behaviour of wild boar (Cutini et al. 2013). The yield values of oak (*Quercus robur* and *Q. petraea*) per year (from 1 July until 30 June) for Lower Saxony were transformed into three classes indicating the condition of the mast crop. Excellent mast crop demonstrates > 70 % of the maximum possible harvest, moderate mast crop condition indicates 10-69 % of the maximum possible harvest, and failure is defined as < 10 % of the maximum possible harvest (Keuling et al. 2013b). Mast crop conditions were excellent in 2003, 2006 and 2009 whereas 2004, 2008, 2010 and 2014 were years of failure in acorns.

Each sampled individual is associated with a hunting index (HI) of the administrative district where it was collected. The HI is defined as the hunting bag per km² and as data were mainly with the same hunting technique, we expect that effort was the same in each year. From 2003 until 2005, the HI increased but decreased

in 2006 to a value of about 1.2. In the next years, the HI increased and decreased again in 2011 to a value of about 1.5, followed by a peak of 2 in 2012 and a low level in 2013 and 2014 (Appendix IV).

To be able to analyse the effect of wild boar density on the timing of birth, estimations of the abundance of wild boar in the study area are necessary. Unfortunately, the expense and effort required to estimate true population densities are prohibitive (Acevedo et al., 2007). Data required by trap cam method (Rowcliffe et al. 2008) are only partly available for the study area so that the HI has to be taken as an indication for the population density of wild boar. The HI is regarded as an appropriate measurement to obtain valid results about the wild boar density (Acevedo et al. 2007).

Statistical analysis

For all tests, significance was assumed when $p < 0.05$. All analyses were performed using R software version 3.1.2 (R Core Team 2014).

To test if birth dates were uniformly distributed throughout the year, birth dates were classified into four periods, autumn (September–November), winter (December–February), spring (March–May) and summer (June–August). The existence of differences in the frequency of events per period was tested by applying a Chi-square test (Lombardini et al. 2014; Manly et al. 2002).

Two different methods were used to calculate the birth date. One based on foetal age (BFA), and the other calculation was based on the hunted piglets (BHP). BFA was calculated according to the formula of Henry (1968): $y = 23.4349 + 0.3230x$, where y is the age of the foetus in days and x the crown–rump length in millimetres. A pregnancy period of 115 days is assumed (Briedermann 2009). Thus, the birth date was calculated by 115 days - $(23.4349 + 0.3230x)$ + hunted date. From 444 wild boar sows (of all age classes) the birth date could be calculated with this method, whereof 116 were piglets.

For the BHP, the birth date was calculated based on the hunted date and the age in months of the shot piglets. Due to increasing inaccuracy with increasing age, only piglets

were taken into account. With this method, the sample size was 1682 (piglets). By the first method summer birth could only be detected partly as samples were only taken from pregnant animals from October until February (Appendix V). Therefore, the second calculation method (BHP) was used to analyse which factors had an impact on summer birth.

To find out which variables influence the time of birth based on BFA calculation, general linear mixed models (GLMM) with restricted maximum likelihood estimation (Zuur et al. 2010) were performed with the package nlme (Pinheiro et al. 2014). The administrative district as well as the hunting year were included as random factors because data collected from individuals from the same year and administrative district may be more dependent on each other.

Since there were correlations between the different parameters (food, weather, hunting, and litter size), and because we are interested in the effect of all of them on the timing of birth, four models were performed.

Maternal body weight and age correlated with each other and could not be put together in one model. Body weight and age class were included in the models as interaction with mast availability and weather conditions, respectively, as we expected different response of weight and age on those environmental factors. The Akaike Information Criterion (AIC) (Burnham and Anderson 2002) was used to decide which physiological parameter should be included in which model (mast or weather conditions).

The third model included the HI from the previous hunting year compared to the year where the samples were collected. This indicates the effect of hunting on the timing of birth. Furthermore, the HI from the same hunting year compared to the year where the data were collected indicates the effect of population density on the timing of birth.

The last model only included the litter size as maternal condition and litter size correlated (unpublished, Frauendorf et al.) and, therefore, did not show reliable results when they were included in one model.

For all models, interaction terms were tested

but are not reported as they were not significant.

Explanatory variables were checked for correlations between each other. If a Pearson correlation was larger than 0.7 or the variance inflation factor (VIF) was larger than 3, one of the two variables was excluded from analysis (Zuur et al. 2010). AIC was used to drop each explanatory variable in turn until the best fitting model was reached. We checked for normality and homogeneity by visually inspecting the plots of residuals against fitted values of the best fitting model. Models were also validated for Cook's distance and temporal autocorrelation. According to Zuur et al. (2010) data transformation should be avoided as long as possible because heterogeneity is interesting ecological information that should not be thrown away. Variance structures for heterogeneity were incorporated in the models. Based on the AIC, the best fitting variance structure was chosen (Zuur et al. 2010).

We expected that harsh weather conditions (Melis et al. 2006) at main birth time (February until April), which are defined by a high number of frost days in interaction with high precipitation (mm) in February, March, and April, would lead to higher juvenile mortality and, therefore, increase the chance of birth in summer. The dependent variable was a binary value: 0 indicating that the piglet was born in the month November until May: 1 indicating that the piglet was born in summer (June until

August). Due to correlations ($p > 0.7$) between precipitation in February and March as well as between the number of frost days of February and April, only frost days and precipitation of the month February were included in the model. In total, 1682 birth dates of hunted piglets (BHP) were analysed.

A generalized linear mixed model (GZLMM) was performed with the package lme4 (Bates et al. 2014). Random structures were added for the same reason as mentioned above. Since the dependent variable was binary, a generalized model was used. In addition, the cloglog-link function was used because the data consisted of considerably more zeros ($N=1619$) than ones ($n=63$) (Zuur et al. 2010). The model was validated based on temporal independence and Cook's distance. Checking or correcting for over -or underdispersion was ignored as it cannot occur in a Bernoulli GLM (Zuur et al. 2010).

Results

Birth dates of piglets were not evenly distributed throughout the year by doing a calculation based on foetal age (BFA) (chi-square test: $\chi^2=243.5172$, $df=3$, $n=116$, $p<0.001$) as well as on hunted piglet age (BHP) (chi-square test: $\chi^2=2424.326$, $df=3$, $n=1682$, $p<0.001$). Both showed a peak in spring, where 87.1 % (BFA) and 75.2 % (BHP) of the piglets were born. Based on hunted piglet calculation,

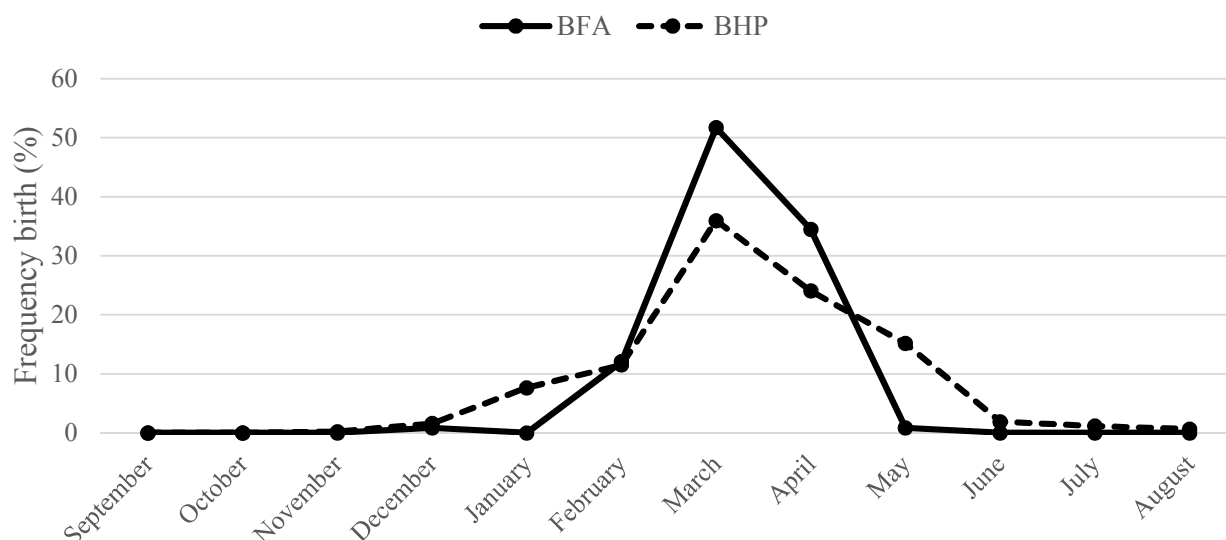


Figure 1: Time of birth based on two different calculation methods: BFA is based on the age of foetuses of piglet sows ($n=116$), BHP is based on the age of hunted piglets (BHP) ($n=1682$). See material and methods for further explanations.

more wild boar were born in summer (3.7 % vs. 0%) and winter (20.8 % vs. 12.9 %) compared to the foetal age calculation (Figure 1).

Several variables influence the birth date of offspring in wild boar (Table 1).

an earlier birth date (GLMM: $z=-2.822662$, $n=444$, $p<0.01$). No relationship was found between the HI and the piglet birth date (GLMM: HI previous year: $z=-0.939955$, $n=444$, $p=0.6082$, HI same year: $z=-0.939955$,

Table 1: Overview of variables influencing the birth date of wild boar offspring obtained by four general linear mixed models. See material and methods for further explanation. Piglets ($n=116$) are used as reference in model 1 as well as mast crop failure ($n=125$) in model 2. Temperature is for the autumn preceding birth. Estimates, standard error (SE) and z-values of the best fitting model are shown. Significance is indicated in the following way: NS=not significant, *= $p<0.05$, **= $p<0.01$, ***= $p<0.001$.

Model	Variable	Estimate	SE	z value	Significance	Effect
1	Yearlings ($n=206$)	-19.31735	2.533823	-7.623795	***	Earlier parturition compared to piglets
	Adults ($n=122$)	-14.23196	2.732875	-5.207688	***	Earlier parturition compared to piglets
	Temperature in autumn ($n=444$)	-2.91672	0.939955	-3.103039	**	Earlier parturition with higher temperature
2	Maternal body weight ($n=444$)	-0.45978	0.075422	-6.096152	***	Earlier parturition with higher weight
	Moderate mast crop condition ($n=177$)	-2.61473	2.781758	-0.939955	NS	No effect
	Excellent mast crop condition ($n=142$)	-8.24957	2.901205	-2.843498	**	Earlier parturition compared to mast failure
3	HI from previous hunting year ($n=444$)	-1.42566	2.758502	-0.5168	NS	No effect
	HI from same hunting year ($n=444$)	4.36978	3.540956	1.234068	NS	No effect
4	Litter size ($n=444$)	-1.64549	0.582956	-2.822662	**	Earlier parturition with higher litter size

Best fitting models (Appendix VI) showed that the offspring of yearlings (GLMM: $z=-7.623795$, $n=206$, $p<0.001$) and adults (GLMM: $z=-5.207688$, $n=122$, $p<0.001$) were born earlier than the offspring of piglets (Figure 2). Earlier birth date occurred when there was a higher temperature in autumn (GLMM: $z=-3.103039$, $n=444$, $p<0.01$) and higher a maternal body weight (GLMM: $z=-6.096152$, $n=444$, $p<0.001$). Better food availability in terms of excellent mast crop conditions of oak also led to earlier birth dates (GLMM: $z=-2.843498$, $n=142$, $p<0.01$) compared to failure mast years whereas moderate mast conditions did not change the birth date significantly (GLMM: $z=-0.939955$, $n=177$, $p=0.3478$). Also a higher litter size resulted in

$n=444$, $p=0.2246$).

The model concerning summer birth indicates that summer birth occurred more often in wild boar when a higher number of frost days in combination with a higher precipitation in February was measured (GZLMM: $z=3.198$, $n=1682$, $p<0.01$). However, in years where precipitation in February and the number of frost days were very low, the probability that piglets are born in summer was higher (Figure 3). In years where only one parameter was high (precipitation or number of frost days), more piglets were born in the winter/spring period

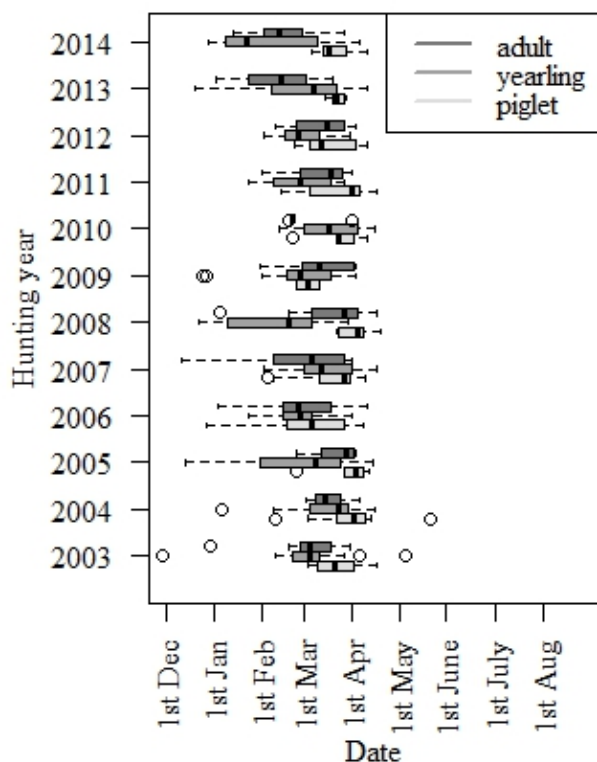


Figure 2: Birth date per age class and hunting year of the studied wild boar population ($n=444$). Box and whisker plots show median (vertical line within box), 25% (interquartile range 1) and 75% (interquartile range 3) percentiles (box) and the range (whiskers), which includes observations up to 1.5 times the interquartile ranges. Circles indicate statistical outliers (observations outside 1.5 times the interquartile ranges).

(precipitation: GZLMM: $z=-3.150$, $n=1682$, $p<0.01$; frost days: GZLMM: $z=-3.571$, $n=1682$, $p<0.001$).

Discussion

Birth frequencies of piglets per month were quite similar with both calculations (BHP and BFA) and showed an uneven distribution throughout the year where birth months ranged from January until May with a peak in March. January and May were underestimated by the BFA calculation whereas March was overestimated. These differences are based on the limited period of data collection with the BFA calculation.

The wide variability in birth months throughout the year in wild boar is contradictory to other ungulates like red deer (*Cervus elaphus*) (Loe et al. 2005) where a high degree of synchrony is reported. For example in roe deer (*Capreolus capreolus*) in Norway,

80 % of birth events occurred within 24 days on inland sites (Linnell and Andersen 1998). An ecological reason for synchronous reproductive behaviour in group-living species is to maximize reproductive success by reducing juvenile mortality by predation. Offspring of synchronous breeders may experience lower predation rates as groups of adult breeders may collectively defend their young from predators or “swamp” the predator population by a mass occurrence of prey in the vulnerable stage, and thus, reducing the fraction of prey taken by the predator (Ims 1990). This hypothesis is probably not applicable in wild boar because they have a different group structure. Family groups of 1-5 sows with their piglets occur most often (Dardaillon 1988). In addition, wild boar is a polytocous species (Gethöffer et al. 2007), so the probability of an individual offspring being attacked by a predator is lower and highly synchronous reproduction is unnecessary. Even though perinatal and postnatal mortality can be quite high in wild boar offspring (Massei and Genov 1997), this seems to be mainly due to the fact that young individuals are more sensitive to environmental variations (Massolo and Della Stella 2006) instead of being attacked by a predator.

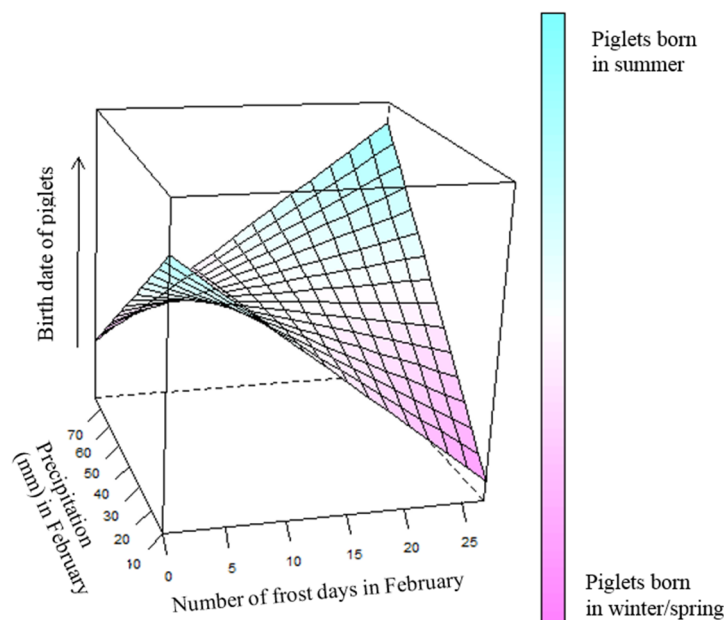


Figure 3: The surface represents the predicted birth time as a function of precipitation and frost days in February obtained by a generalized linear mixed model. At extreme precipitation in combination with extreme number of frost days in February, the probability that piglets are born in summer months (June, July and August) is higher ($n=63$). If only precipitation or number of frost days is high, piglets are mainly born in winter/spring ($n=1619$).

A second reason might be forage conditions that are restricted to a certain period of the year (Post and Stenseth 1999).

For instance, female bighorn sheep (*Ovis canadensis*) that matched the peak in plant growth with the lactation period had a higher fitness (Festa-Bianchet 1988). It could be assumed that the food conditions in the studied population are very good, probably due to agriculture intensification (Geisser and Reyer 2005), and that there is no need for synchronous breeding for wild boar. This also fits with the results from Fernández-Llario and Carranza (2000) who reported shorter reproductive periods under harsher environments such as during drought conditions in summer. Summer precipitation was not reported as it had no effect on the timing of birth in the studied wild boar population, similar to the study of Geisser and Reyer (2005) where rain from April until June did not affect the population density in Switzerland. Probably, no effect was found in comparison with the study of Fernández-Llario and Carranza (2000) because Western Spain is a drier and hotter region than Northern Germany, where rain conditions have a higher effect on the reproductive behaviour of wild boar.

Wild boar adjusted birth date to several factors. Birth date occurred earlier when the sows had a higher age, larger body weight and larger litter size.

Maternal age and body weight have a positive effect on the birth date (Briedermann 2009). As older and heavier animals might have more energy to invest in their offspring (Bieber and Ruf 2005; Geisser and Reyer 2005), they are able to give birth earlier than lighter piglets that just became mature. This was also found in moose (*Alces alces*) in Estonia, where older individuals conceived earlier than younger animals (Veeroja et al. 2013).

In addition, a positive correlation was found between the litter size and maternal body weight and age (Fernández-Llario et al. 1999; Fonseca et al. 2011) so that individuals with larger litter sizes can reach an earlier birth date, even though a larger litter size may need more energy.

Especially in ectotherm animals, temperature and humidity influence the reproductive physiology and population dynamic (Walther et al. 2002). But also studies on large mammals indicated that extreme climate influenced juvenile survival (Milner et al. 1999), fecundity (Post and Stenseth 1999) as well as time of maturation, time of birth and weight at birth (Forchhammer et al. 2001).

We found that earlier parturition occurred when temperature in autumn was higher. Usually, gregarious, placental mammals can synchronize breeding by adjusting the time of oestrus but not the time of birth (Berger 1992). Similarly, Mauget and Pépin (1985) found that wild boar shortened the anoestrus under food abundance in September and October. 71 % of the wild boar sows in the studied population were in heat in autumn (September, October and November). Thus, it might be reasonable that the conception time is also adjusted under favourable temperature conditions.

Berger (1992) demonstrated that female bison (*Bison bison*) in good body condition that mated after the seasonal peak shortened gestation time in order to synchronize births with other females. In wild boar, however, temperature in autumn did not interact with the age classes (and therefore also body weight). Thus, piglets and older sows did not respond in a different way to weather conditions concerning the timing of birth.

Although less pronounced, there is some evidence for a later onset of autumnal phenological events in bird or tree species due to climate change (Walther et al. 2002). However, as climate change scenarios indicate substantial warming in all seasons in the whole of Europe (Giorgi et al. 2004), we might expect that wild boar conditions for earlier conception and parturition will become better for successful and higher reproduction in the future.

It is known that levels of seasonality and inter-annual variability in resource dynamics shape the level of breeding synchrony in ungulates (English et al. 2012), and also in wild boar, numerous studies noted the effect of food condition on the parturition time (Aumaitre et al. 1982; De Vos and Sassani 1977; Fernández-

Llarío and Mateos-Quesada 1998; Maillard and Fournier 2004; Pépin et al. 1987). In general, mammals can adapt conception time and gestation length but not birth time (Berger 1992), which may depend on physiological parameters as mentioned above. Most animals in the studied population conceived in autumn (see above), so this is in line with the time of greatest mast availability. In years of excellent mast condition, wild boar sows are in better physiological condition and can adapt oestrus time to reach an earlier birth (Mauget and Pépin 1985). Piglets born earlier can reach the necessary physical requirements earlier, which control the start of the reproductive cycle (Gamelon et al. 2011; Massei et al. 1996) and take part in the reproduction already before reaching their first year. Matschke (1964) has already reported that the number of females participating in reproduction increased during good mast years. As the frequency of mast years will probably increase further in the future (Övergaard et al. 2007), this may favour early parturition.

Against our expectations, we did not find any interaction between the maternal body weight and the mast condition. We expected that piglets with a lower weight would respond to a higher extent to mast years than yearlings or adults. Probably, the all year round food supply by agriculture intensification (Schley and Roper 2003) makes the wild boar less dependent on mast availability (unpublished, Frauendorf et al.). Thus, we can conclude that there is no age-specific adaptation under different food conditions concerning parturition time.

Intensification of agriculture is recognized as one of the most significant human alterations to the global environment (Matson et al. 1997), and as wild boar make use of agricultural food in summer (Keuling et al. 2009), the food situation for wild boar in Germany is favourable for early parturition and the high reproduction rate of wild boar.

Further, we should take into account that there might be differences in the response of sows with different habitat use patterns during summer (Keuling et al. 2009). Probably sows that remain in the forest during the summer are

more dependent on mast years than “field sows” that shift their home range entirely from the forest to the fields in the summer.

We could not find any effect of hunting on the timing of birth. In contrast, Gamelon et al. (2011) found in a long-term (22 years) monitoring in France that a larger hunting bag led to earlier parturition. This might result from different time length of the study period as well from different methodology. They estimated birth date by using the weight of a juvenile, assuming birth weight equals 1 kg and using a linear model to predict individual growth rate over the first months of life. In addition, they used hunting pressure in two different periods where the difference between the weak (1.6 individuals/km²) and high hunting pressure (4.2 individuals/km²) was much higher than in our studied population.

The probability that wild boar are born in summer was higher when both the number of frost days and the amount of precipitation in February were extremely high or low. This is an indication that, as expected and described in the introduction, under extremely harsh weather conditions (wet and cold) at the time of birth, preweaning mortality (Akos and Bilkei 2004) so that sows reproduce a second time in the year (Briedermann 2009). In addition, if weather conditions are really good in February, meaning warm and dry, there is the possibility for a second litter because sows might have more energy and a better physiological condition to be able to invest in it. This is similar to observations found in farmed fallow deer (*Dama dama*) where older does, who have higher energy in terms of body weight, showed more oestrus cycles than younger ones (Asher 1985). Cold and dry weather, however, do not have a serious effect on healthy domestic pigs (Akos and Bilkei 2004), which might be similar in wild boar juveniles so that sows only mated one time in autumn.

A possible methodological reason for the low percentage of piglets born in summer is the fact that reproductive tracts from piglets which were younger than 5 months were hardly taken during data collection (only two piglets with an age of 3 month) as the chance of pregnancy was

minimal. Since most data from piglets were collected in November (37.9 %), December (29.8 %) and January (29.4 %), animals born in the summer period could be missed. Another possible ecological reason might be that spring starts earlier in the year (Walther et al. 2002), which changes the timing of spring activities like earlier breeding in amphibians (Blaustein et al. 2001) and birds (Crick and Sparks 1999) so that weather conditions are less harsh in February. Probably, several decades ago or in mountainous areas (Durio et al. 1995; Moretti 1995), where weather conditions are more extreme, summer birth occurred more often in wild boar.

Conclusion

Sows in better physiological condition (higher body weight) showed earlier parturition. Since average dressed maternal body weight increased in piglets, yearlings and adults with 12 %, 13 % and 11 %, respectively (Lustig 2015) during the study period, we can expect an increase in reproduction in wild boar populations in the future. In addition, also favourable weather conditions in autumn as well as higher frequency of excellent mast conditions, working indirectly via maternal body mass, will lead to increased reproduction as piglets are born earlier and can reach maturity in the same year.

We also can conclude that due to climate change, we have to expect earlier parturition dates in wild boar in the future, which might increase the number of piglets that are able to participate in reproducing in the same years as they are born. At the moment, only a small percentage of wild boar piglets are born in summer. As we can expect from weather scenarios that spring will start earlier in the future, it could be that summer birth will occur less frequently in the future due to less harsh weather conditions. However, summer birth events could also increase since the weather conditions in February will be so good that they will have enough energy to invest in a second litter. Therefore, we need further research to investigate, which scenario is more likely in order to adapt probably also guidelines for

hunting in the summer months.

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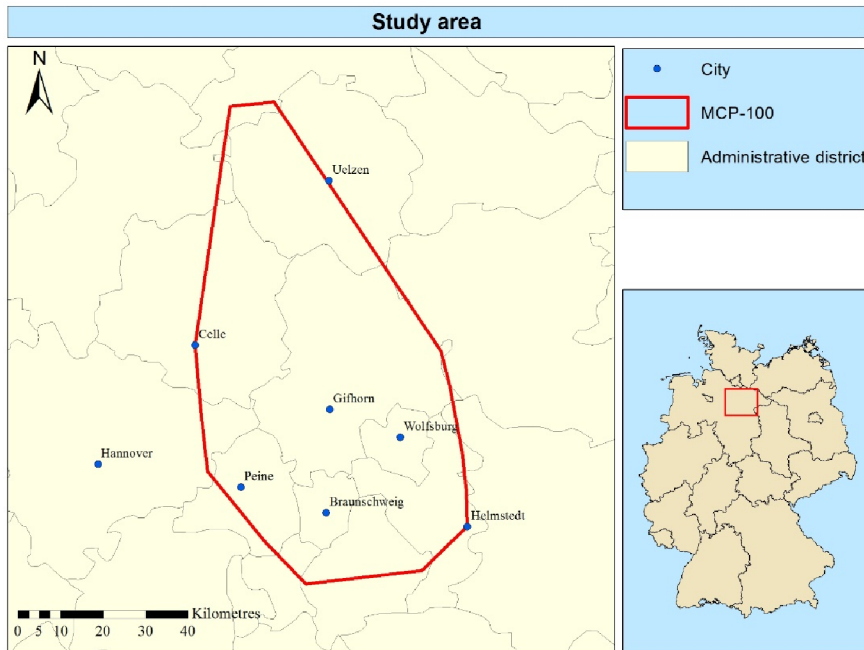
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Appendix

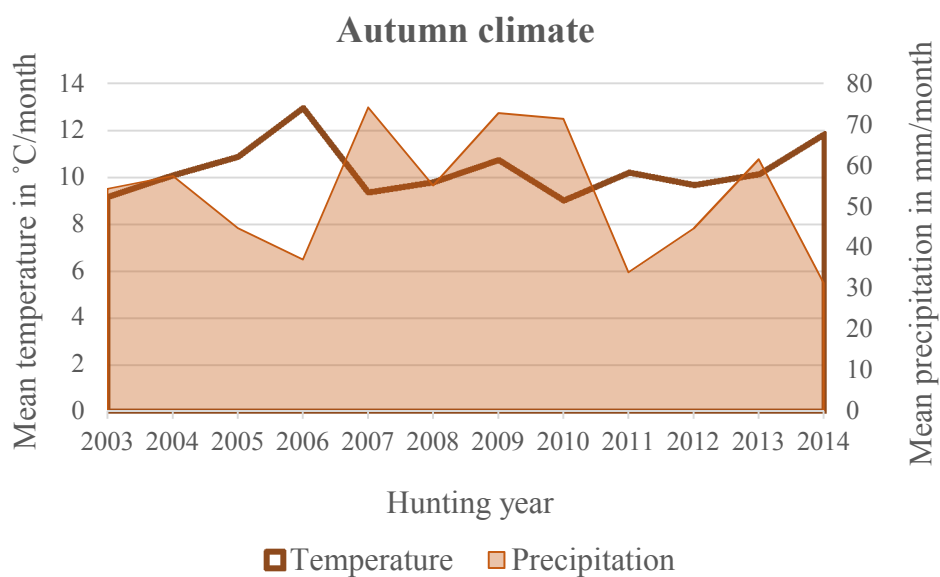
Appendix I:

Map of the study area in Northern Germany, showing a MCP-100 around the sampled locations, some cities as orientation and the administrative districts in the background.



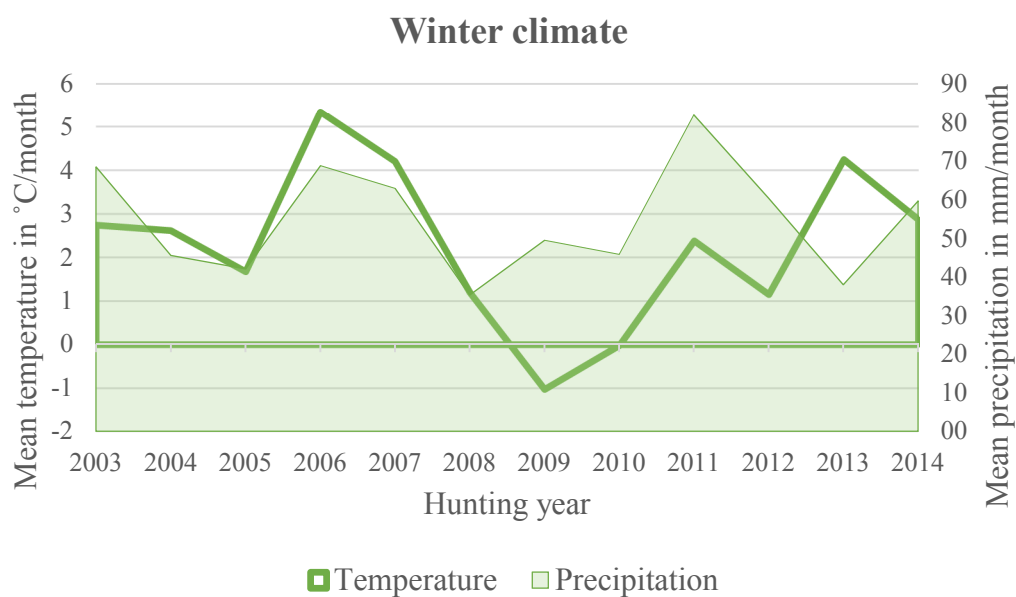
Appendix II:

Mean temperature in °C/month and mean precipitation in mm/month in autumn in the study area of the 12 wild boar hunting seasons in Germany.



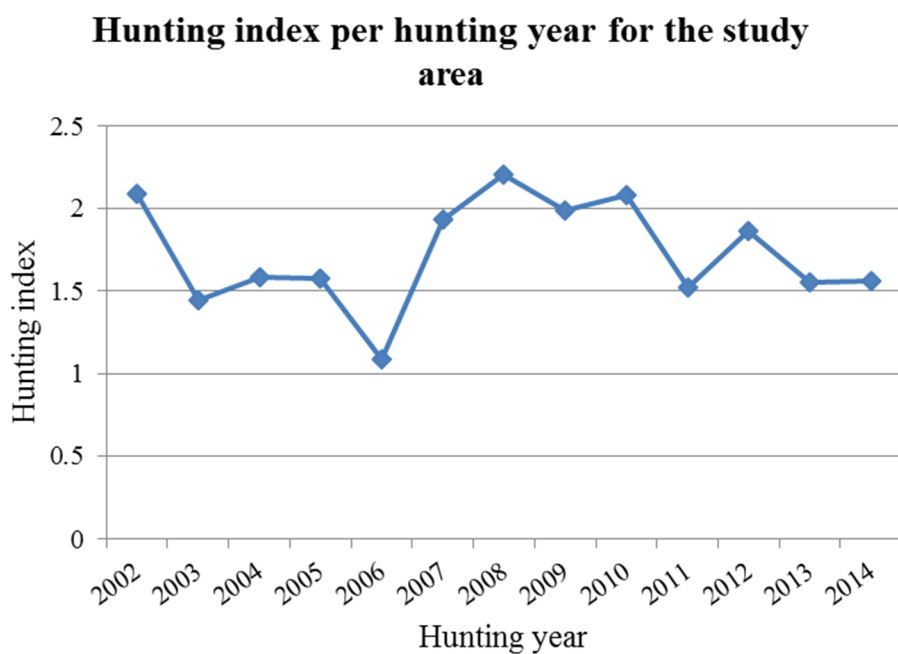
Appendix III:

Mean temperature in °C/month and mean precipitation in mm/month in winter in the study area of the 12 wild boar hunting seasons in Germany.



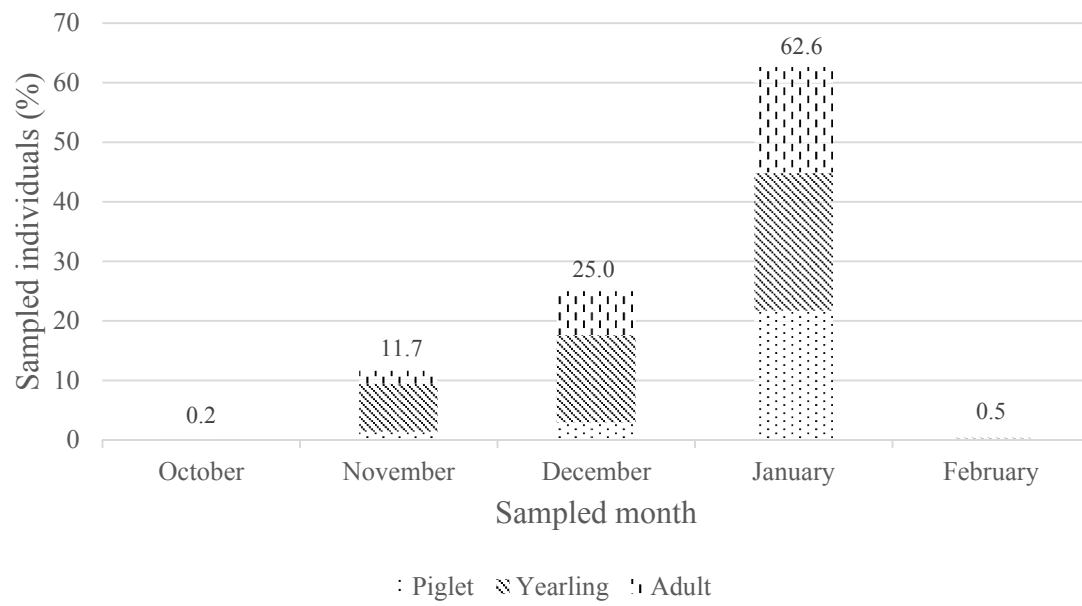
Appendix IV:

Hunting index (HI=hunting bag/km²) for the study area for the 12 wild boar hunting seasons.



Appendix V:

Distribution of sampled pregnant individuals (in %) per month.



Appendix VI:

Model 1: Weather conditions and age class included as independent variable. Bold models indicate best fitting model of the round.

Round	Dropped independent variable	Model with included independent variables	AIC
1	None	AgeClass+TemperatureAutumn+TemperatureWinter+PrecipitationWinter	4064.599
	AgeClass	TemperatureAutumn+TemperatureWinter+PrecipitationWinter	4115.518
	Temperature Autumn	AgeClass+TemperatureWinter+PrecipitationWinter	4071.036
	Temperature Winter	AgeClass+TemperatureAutumn+PrecipitationWinter	4063.812
	Precipitation Winter	AgeClass+TemperatureAutumn+TemperatureWinter	4061.248
2	None	AgeClass+TemperatureAutumn+TemperatureWinter	4061.248
	AgeClass	TemperatureAutumn+TemperatureWinter	4112.701
	Temperature Autumn	AgeClass+TemperatureWinter	4068.828
	Temperature Winter	AgeClass+TemperatureAutumn	4060.463
3	None	AgeClass+TemperatureAutumn	4060.463
	AgeClass	TemperatureAutumn	4112.524
	Temperature Autumn	AgeClass	4069.344

Model 2: Mast conditions and maternal weight included as independent variable. Bold models indicate best fitting model of the round.

Round	Dropped independent variable	Model with included independent variables	AIC
1	none	Mast+Weight	4080.238
	Weight	Mast	4110.662
	Mast	Weight	4092.365

Model 3: Hunting index (HI=hunting bag/km²) of the previous and same hunting year included as independent variable. Bold models indicate best fitting model of the round.

Round	Dropped independent variable	Model with included independent variables	AIC
1	none	HIPreviousYear+HISameYear	4113.395
	HISameYear	HIPreviousYear	4117.262
	HIPreviousYear	HISameYear	4115.524